Road and off-road vehicle design

Mechanical Engineering

Automotive and Motorsport Engineering, M.Sc.

Assignments Summary

Group:

Pietro Nardon

Cristiano Riganti

Gianluca Meli

February 7, 2025

Contents

1 Assignment 1: General performance of a motor vehicle

1.1 Abstract

The "General Performance of a Motor Vehicle" assignment is based on the derivation of the two following quantities:

- Maximum speed
- Acceleration performances

In this context, the following vehicle has been analyzed: Citroen C3 PureTech 110 (man. 6) 2024.
Two different configurations has been considered: a lighter and a heavier one.

In the following sections an almost purely analytical approach has been used (apart from some experimental coefficients) and some quantities has been estimated without any testing, often exploiting the available data on the *Automobile-catalog* website.

Finally, these results has been obtained:

- Maximum speed (light vehicle): about 188 km/h
- Maximum speed (heavy vehicle): about 184 km/h
- 70 - 120 km/h acceleration (light vehicle): about 12 s
- 70 - 120 km/h acceleration (heavy vehicle): about 16 s

1.2 Introduction

The aim of this project is to compute the maximum speed and the acceleration performances of a motor vehicle, through the manipulation of some available data and the analytical derivation of certain quantities.

The chosen vehicle (Citroen C3 PureTech 110 - man. 6 - 2024) is a 5-door hatchback, petrol and 6-gears manual powertrain car, whose data are presented in the following chapters.

In order to characterize the general performance, all the possible resistance forces need to be calculated for all the possible speed values, namely the weight-related resistance (sloped road), the rolling resistance and the aerodynamic resistance. The total resistance power is the sum of the resistance forces (depending on vehicle speed v) multiplied by the associated speed.

Gear ratios and efficiency coefficients have been exploited to compute the required torque/power at the engine.

In the following sections all the quantities are computed and all the related data are presented.

1.3 General performance calculations

1.3.1 Request 1 - Car Choice

For this analysis, the following vehicle (Fig. 1) has been chosen:

- Citroen C3 PureTech 110 - man. 6 - 2024

Figure 1: Citroen C3 PureTech 110 - man. 6 - 2024

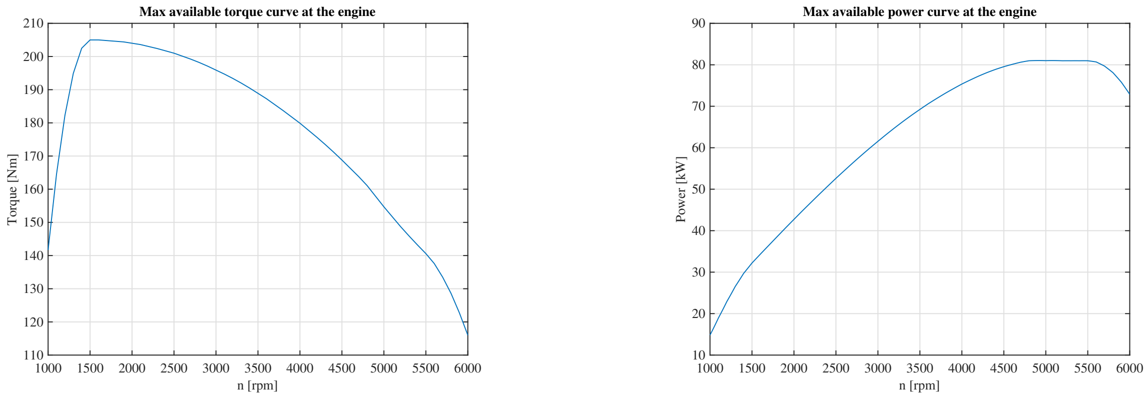
This is a petrol-powered, 5-door hatchback (segment B) car, characterized by a 1199 cm^3 engine, with 81 kW of power, 205 Nm of torque and a 6-speed manual powertrain, sold up to April 2024 in Europe

1.3.2 Request 2 - Main vehicle features

By exploiting the technical sheets available at *www.automobile-catalog.com*, the following data have been acquired:

- Curb mass: $m = 1090\,kg$
- Wheelbase: $L = 2.539\,m$
- Maximum cross-sectional area: $A_f = 2.14\,m^2$
- Drag coefficient: $C_d = 0.31$
- Tires: 205/50 R 17
- Front Wheels Drive

The following engine characteristics (Fig. 2) have been plotted by using the data available on the cited website.



(a) Torque at the engine

(b) Power at the engine

Figure 2: Engine characteristics

1.3.3 Request 3 - Transmission ratios and efficiencies

The gearbox and final drive transmission ratios, from the data sheets, are:

τ_I	τ_{II}	τ_{III}	τ_{IV}	τ_V	τ_{VI}	τ_{final}
3.455	2.048	1.29	0.975	0.745	0.574	4.29

Table 1: Gear ratios

The related efficiencies have been estimated as follows:

η_I	η_{II}	η_{III}	η_{IV}	η_V	η_{VI}	η_{final}
0.90	0.92	0.94	0.95	0.96	0.97	0.95

Table 2: Gears efficiency

1.3.4 Request 4 - Resistance and engine powers as function of speed

Two different configurations are considered:

- Light: $M_l = m_{curb} + m_{person} + m_{luggage,light} + 0.5 \cdot m_{fuel,full} = 1182\ kg$
- Heavy: $M_h = m_{curb} + 5 \cdot (m_{person} + m_{luggage,heavy} + m_{fuel,full}) = 1549\ kg$

Where:

m_{curb}	m_{person}	$m_{luggage,light}$	$m_{luggage,heavy}$	$m_{fuel,full}$
$1090\ kg$	$70\ kg$	$5\ kg$	$15\ kg$	$33.75\ kg$

Table 3

For the resistance power calculation, the following additional data are provided:

f_0	f_2	ρ	C_d	A_f
0.01	$6.48 \cdot 10^6 s^2/m^2$	$1.225 kg/m^3$	0.31	$32.14 m^2$

Table 4

The three contributions of resistance force are computed as follows:

$$R_{slope} = P sin(\alpha) \tag{1.1}$$

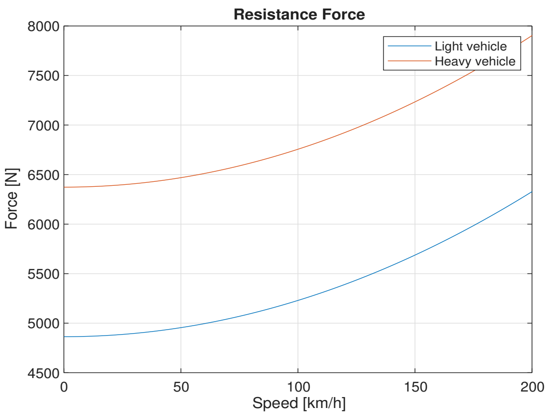
$$R_{rolling} = mg \cdot cos(\alpha)(f_0 + f_2 v^2) \tag{1.2}$$

$$R_{aero} = 0.5 \rho C_d A_f v^2 \tag{1.3}$$

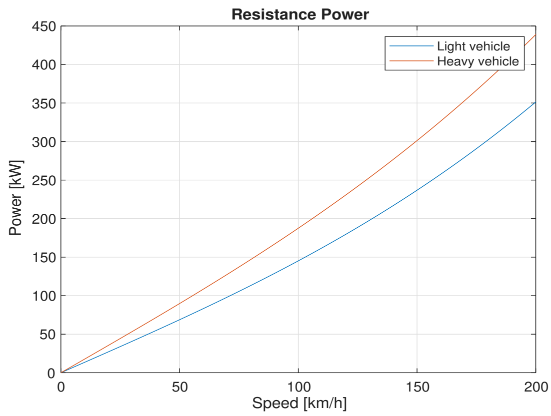
Finally, the total resistance power can be computed as:

$$N_{r,tot} = (R_{slope} + R_{rolling} + R_{aero}) v \tag{1.4}$$

In (Fig. 3) the total resistance force and power are plotted.



(a) Force



(b) Power

Figure 3: Resistance force and power

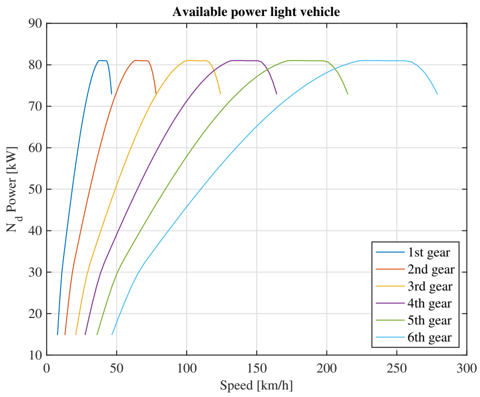
By exploiting the gear ratios of (Tab. 1) and the effective radius computed in (Eq. 1.5)

$$R_e = R_0 - \frac{F_{Z0}}{k_z} \tag{1.5}$$

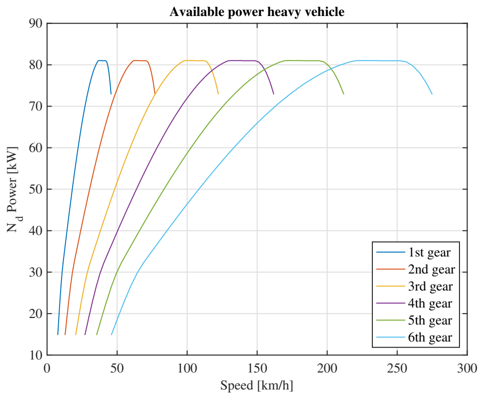
where F_{Z0} is the static vertical load (weight), $k_z = 200000\, N/m$ is the tyre radial stiffness and

$$R_0 = \frac{\frac{w \cdot AR}{100} + \frac{D_{rim}}{2} \cdot 25.4}{1000} \quad [m] \tag{1.6}$$

the unloaded radius, with tyre width $w = 205\, mm$, aspect ratio $AR = 50$ and $D_{rim} = 17\, in$, the engine power (no efficiencies are considered) is plotted in function of the car velocity in (Fig. 4) for the two configurations.



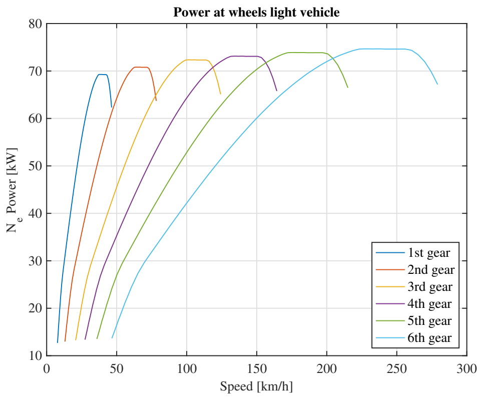
(a) Light configuration



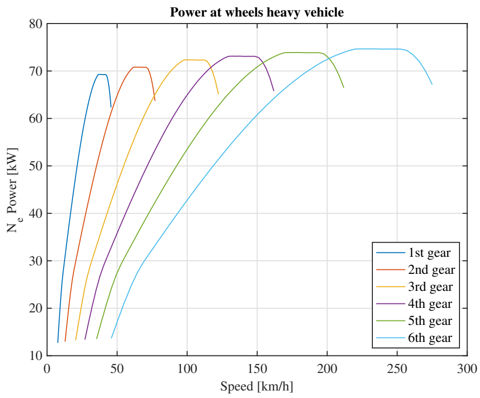
(b) Heavy configuration

Figure 4: Engine power in function of vehicle velocity

By considering also the efficiencies, the available power at the wheels is plotted in (Fig. 5).



(a) Light configuration

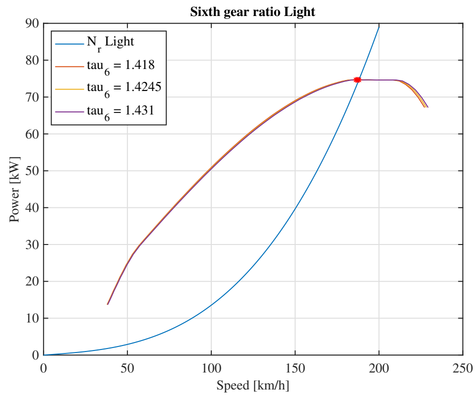


(b) Heavy configuration

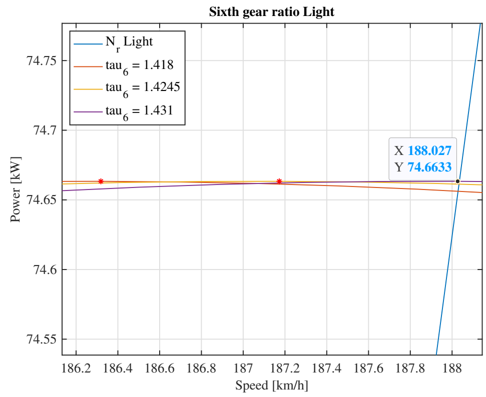
Figure 5: Available power at wheels in function of vehicle velocity

1.3.5 Request 5 - Highest gear ratio

In order to determine the VI gear ratio, a desired maximum speed of about 190 km/h is chosen, following the estimations of the cited website. The idea is to take a average gear ratio value in between the light and heavy configurations, so to obtain a trade-off result. To do that, some τ values have been tried, aiming to maximize the top speed, being it always slightly lower than 190 km/h. On this purpose, the procedure is shown in (Fig. 6), (Fig. 7) and (Fig. 8).
A graphical approach was carried out, changing the values of τ until the maximum of the engine power intersected with the resistance power, the final value was the one bringing the maximum engine power closer to the resistance one.

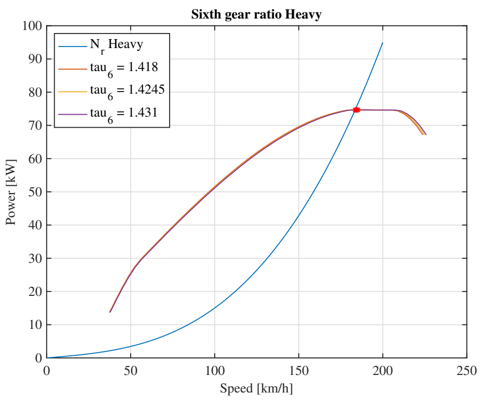


(a) General

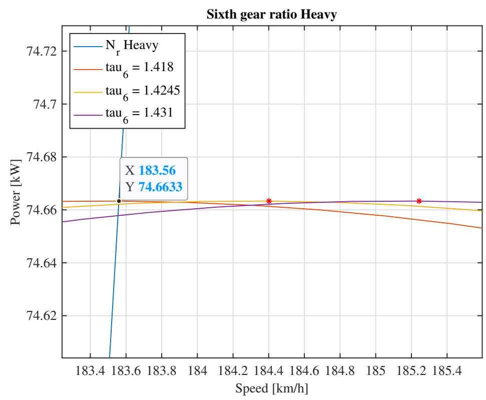


(b) Zoomed

Figure 6: Light configuration

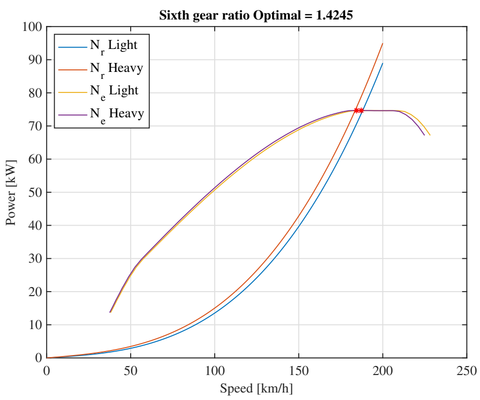


(a) General

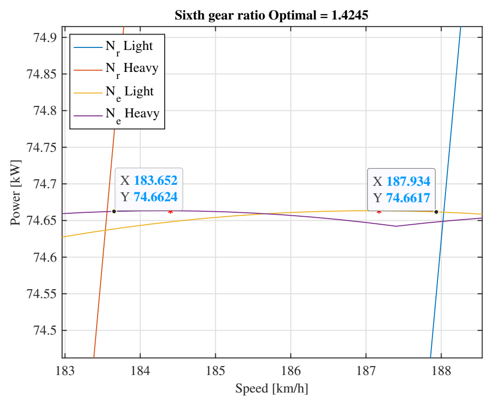


(b) Zoomed

Figure 7: Heavy configuration



(a) General



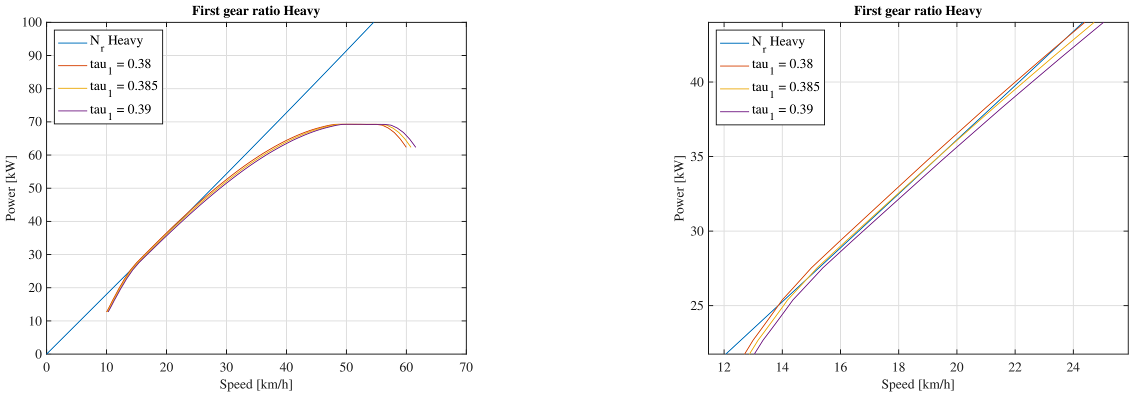
(b) Zoomed

Figure 8: Final choice

Finally, a value of $\tau = 1/1.4245 = 0.702$ is chosen, replacing the original value of 0.574, with a maximum speed of about 188 km/h for the light configuration and of about 183.5 km/h for the heavy one. Possibly, the shorter gear ratio could be partly justified by the choice of the (available) 17 *in* rim instead of the original 15 *in* one.

1.3.6 Request 6 - Lowest gear ratio

In order to determine the I gear ratio, the capability of climbing a 45 % slope needs to be ensured. Operationally the aim is to select τ_I in such a way that the available power curve results to be tangent to the 46 % sloped road resistance curve. Then, a speed between 7 km/h and 12 km/h needs to be verified while running at minimum (1000) rpm in I gear. The 45 % slope condition is considered just for the heavy configuration, corresponding it to the worst possible case. The procedure is shown in (Fig. 9)



(a) General (b) Zoomed

Figure 9: Available and resistance power with 46 % slope

A value of $\tau_I = 1/0.385 = 2.597$ seems to be suggested, which guarantees:

$$v_{min,light} = 10.28\ km/h \qquad v_{min,heavy} = 10.13\ km/h$$

As a demonstration, the curves corresponding to 45 % slope are plotted in (Fig. 10), in which the capability of climbing such a road is verified, even if the use of clutch if starting from null velocity is required.
In heavy configuration, the maximum speed is about 25 km/h.

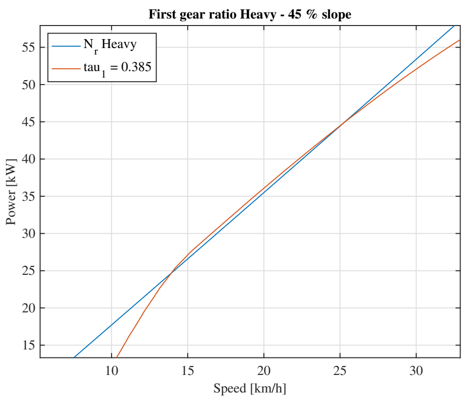
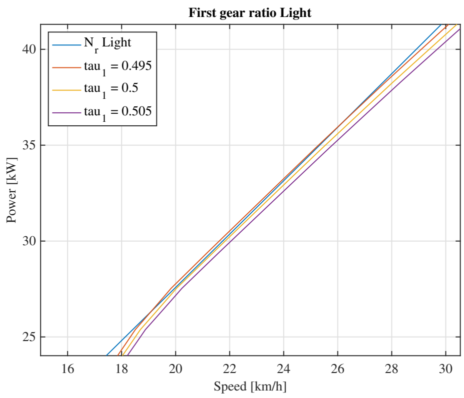
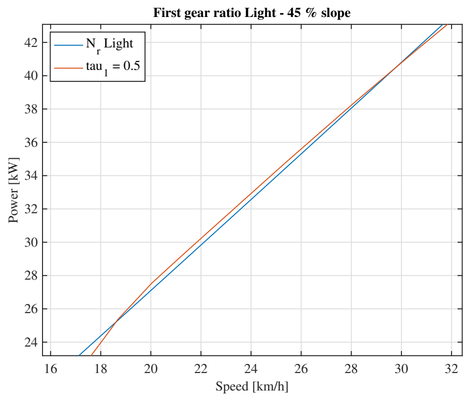


Figure 10: Available and resistance power with 45 % slope

Just for sake of completeness, the procedure is shown also for the light case, but considering the previous results as the only correct ones.



(a) 46 % slope



(b) 45 % slope

Figure 11: Available and resistance powers

The suggested gear ratio would be $\tau_I = 1/0.5 = 2$. However, the two minimum velocities would result to be $v_{min,light} = 13.35\ km/h$ and $v_{min,heavy} = 13.16\ km/h$ (out of bounds) and the final choice would be $\tau_I = 1/0.44 = 2.273$ corresponding to $v_{min,light} = 11.75\ km/h$ and $v_{min,heavy} = 11.58\ km/h$.

1.3.7 Request 7 - Intermediate gear ratios

The II, III, IV and V gear ratios are found through the following geometric progression (Eq. 1.7), which ensures constant engine angular speed variation while changing gear.

$$\tau_{i+1} = \tau_i \left(\frac{\tau_{min}}{\tau_{max}} \right)^{\frac{1}{n-1}} \tag{1.7}$$

where $n = 6$ is the number of gear ratios.

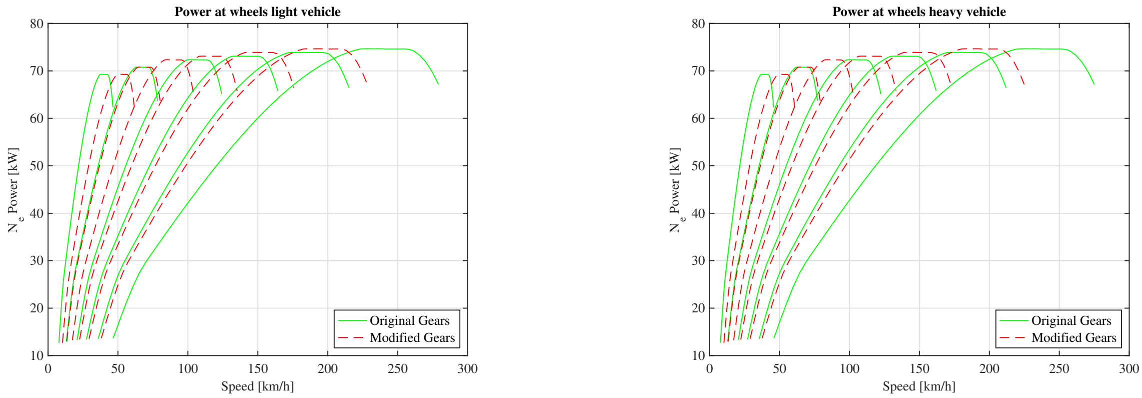
The final gear ratios are summarized in (Tab. 6) and plotted in (Fig. 12)

τ_I	τ_{II}	τ_{III}	τ_{IV}	τ_V	τ_{VI}	τ_{final}
3.455	2.048	1.29	0.975	0.745	0.574	4.29

Table 5: Old gear ratios

τ_I	τ_{II}	τ_{III}	τ_{IV}	τ_V	τ_{VI}	τ_{final}
2.597	1.999	1.539	1.185	0.912	0.702	4.29

Table 6: New gear ratios



(a) Light configuration

(b) Heavy configuration

Figure 12: Available power at wheels - Old vs new gear ratios

1.3.8 Request 8 - Acceleration from $70\,km/h$ to $120\,km/h$

For this final request, the time needed to accelerate the car from $70\,km/h$ to $120\,km/h$, using the VI gear, is computed by adopting the hypothesis of constant acceleration within subsequent values of the discretized velocity vector. In particular, this procedure has been adopted, for both the light and the heavy configurations:

1. The velocity vector corresponding to the VI gear is considered between $70\,km/h$ and $120\,km/h$;
2. The resistance and available forces within this interval are considered and the useful force is computed by subtracting the two;
3. The acceleration for each interval is computed by dividing each useful force value with the considered vehicle mass and the average between two subsequent values is considered to be representative of the constant acceleration of the interval itself;
4. The time of a single interval is computed as $t_i = \frac{V_{i,final} - V_{i,initial}}{a_i}$
5. The total time is computed as the sum of all the single interval time values.

Finally

$$t_{light} = 11.96\,s \tag{1.8}$$

$$t_{heavy} = 16.14\,s \tag{1.9}$$

1.4 Conclusions

As commented in the specific sections, the obtained gear ratios result in a longer first gear and a shorter sixth gear, with a consequent "shrieked" velocity-power diagram (Fig. 12). Especially for the sixth gear, the shorter ratio which has been obtained may be related to the choice of the available 17 *in* rim instead of the original and of reference 15 *in* one. The final general performances are summarized below in (Tab. 7).

	Maximum Velocity	**70 - 120 km/h Acceleraion**
Light Vehicle	$188\,km/h$	$11.96\,s$
Heavy Vehicle	$183.5\,km/h$	$16.14\,s$

Table 7: Velocity and acceleration results

With

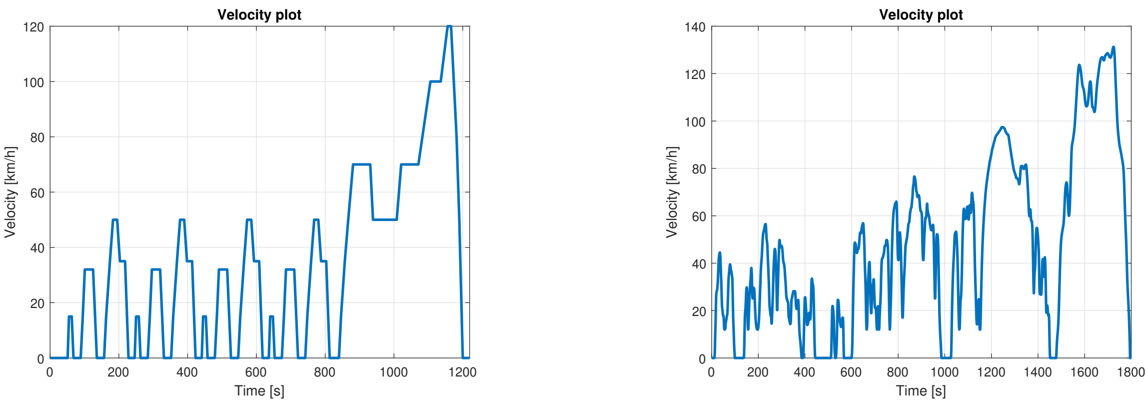
τ_I	τ_{II}	τ_{III}	τ_{IV}	τ_V	τ_{VI}	τ_{final}
2.597	1.999	1.539	1.185	0.912	0.702	4.29

Table 8: Final gear ratios

2 Assignment 2: Consumption analysis of a motor vehicle

2.1 Abstract

Aim of the assignment "Analysis of consumption" was to analyse and asses the fuel consumption of a motor vehicle, using a global, harmonized standard for a light-duty road vehicle. The studied vehicle was the same used in (Sec. 1.3.1), that is a 2024 Citroen C3 PureTech 110 hp, (Fig. 1), with a six gears manual gearbox, 1199 cm^3 petrol engine. The main objectives were to compute the fuel consumption of the car to while running two different test cycles and compare it with the values provided by the manufacturer. The velocity profiles of the cycles were extracted with `Simulink` blocks and then a `Matlab` script was created to perform all the needed plots and calculations, using the formula provided by the theory and by the assignment text. The two cycles had different velocity profiles: the NEDC cycle is equal for each vehicle, and it is shown in (Fig. 13a), while the WLTP cycle differentiates the velocity profiles depending on the specific power output of the vehicle. Our vehicle ended up in class 3, since its power to unladen mass ratio was $74.13W/kg > 34\,W/kg$, therefore its velocity profile was described in (Fig. 13b).



(a) NEDC Velocity profile

(b) WLTP Velocity profile

Figure 13: Velocity profiles

The main results found are reported in (Tab. 9) comparing the two different test cycles. The declared consumption from the manufactured were found in *WLTP info website* and *Automobile-catalog website*.

NEDC		WLTP	
$\dot{m}_{avg}$	$3.9423 \cdot 10^{-4} kg/s$	$\dot{m}_{avg}$	$5.827 \cdot 10^{-4} kg/s$
m_{tot}	$0.4810\,kg$	m_{tot}	$1.0489\,kg$
L_{tot}	$0.6499\,L$	L_{tot}	$1.4175\,L$

Computed	$5.896\,L/100km$	Computed	$6.092\,L/100km$
Manufacturer	$4.4\,L/100km$	Manufacturer	$5.4\,L/100km$

Table 9: Main results of the analysis

2.2 Introduction

To compute the fuel consumption, the total power required by the engine P_{engine} was needed, and in order to find it the resistant power P_{res} and engine efficiency η_{engine} were required.
After extracting the time, velocity and acceleration profiles for both cycles it was possible to compute the resistant power P_{res}, which is the power needed to move the vehicle at the required speed and acceleration, as shown in the following (Eq. 2.1) for each time instant of the simulation. To compute the resistance force R_{tot} the same algorithm and values described in (Sec. 1.3.4) where used, adapting it to take into account also intertia forces.
Then, to compute the power required by the engine P_{engine}, the resistant power was divided by the final drive efficiency and by the selected gear efficiency, as presented in (Eq. 2.3). The gears were selected using the table (Tab. 10).

$$R_{tot} = R_{rolling} + ma + R_{aero} \tag{2.1}$$

$$P_{res} = R_{tot} v \tag{2.2}$$

$$P_{engine} = \frac{P_{res}}{\eta_{gear,j}\eta_{final}} \tag{2.3}$$

v [km/h]	Gear change
15	$I \rightarrow II$
35	$II \rightarrow III$
50	$III \rightarrow IV$
70	$IV \rightarrow V$
100	$V \rightarrow VI$

Table 10: Gear change

To compute the power needed from the fuel, to subsequently derive the fuel consumption, the engine power P_{engine} was needed to be divided by the engine efficiency η_{engine}. To compute η_{engine} it was required to enter a graph where, at a given engine rotational speed and specific work, η_{engine} could be extracted. Starting from the efficiency map for a petrol engine, in (Fig. 14), it was scaled in order to reach the maximum specific work which characterized our vehicle, the rotational speeds instead were already equal to the vehicle's ones, so they were not scaled. It was then sampled into `Matlab` resulting in (Fig. 15), to be interpolated into the needed values at each rotational speed and specific work.

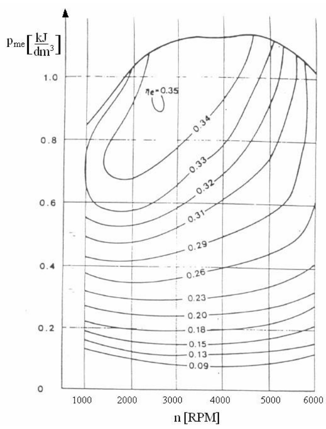


Figure 14: Original efficiency map

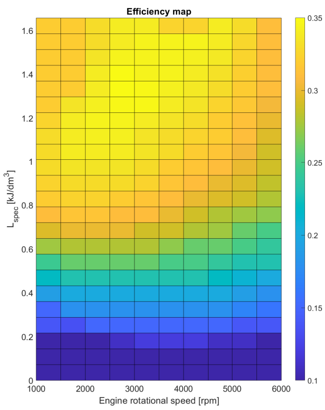


Figure 15: Scaled efficiency map

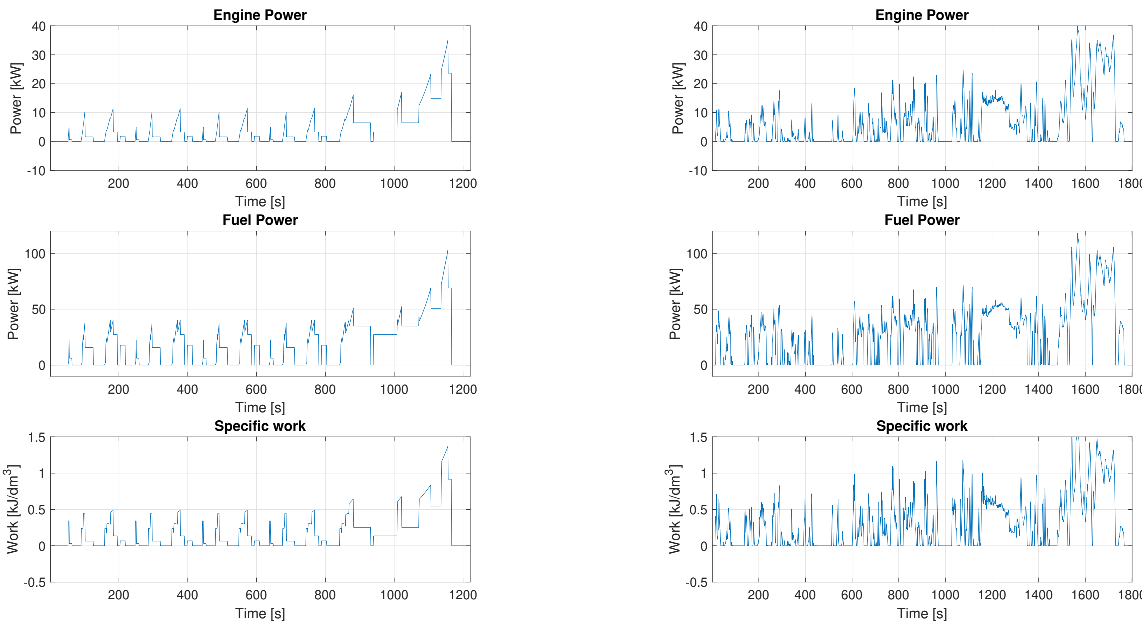
The specific work in (Eq. 2.4), where n_{engine} is the engine rotational speed and C is the engine capacity.

$$L_{spec(i)} = \frac{2 \cdot 60 \cdot P_{engine(i)}}{n_{engine(i)} C} \tag{2.4}$$

2.3 NEDC versus WLTP Cycle

2.3.1 Request 1: Powers computations

The required power at the engine P_{engine}, specific work L_{spec} and fuel power P_{fuel} were computed using the formula expressed in (Sec. 2.2), assuming that while the vehicle is decelerating the power required from the engine is null. Moreover since the vehicle is fitted with a "start and stop" system, in idle condition the engine power is again null. These quantities are plotted in (Fig. 16), and in both cases is easy to distinguish when the car is being tested for a urban environment (in the beginning of the cycle), resulting in continuos cycles of accelerations and braking at relatively low speed, and a highway environment (at the cycle end), resulting in a longer time intervals at higher speed, which request an overall higher power.



(a) NEDC Cycle

(b) WLTP Cycle

Figure 16: Powers and specific work

Taking a closer look to P_{fuel} it was computed by dividing the engine power with the engine efficiency. In order to do that the sampled efficiency described and showed in (Sec. 2.2) was interpolated in order to match the dimensions of the `Matlab` vector describing P_{engine}, effectively creating a lookup table. After computing the fuel power it was possible to derive the mass of fuel consumed at each time interval i in both cycles, using the data provided in (Fig. 17) and the formula (Eq. 2.5).

$$m_{fuel(i)} = \frac{P_{fuel(i)} \cdot \Delta t_{(i)}}{Hi} \tag{2.5}$$

ρ	$0.740\, kg/m^3$
Hi	$43.3 \cdot 10^3\, kJ/kg$

Figure 17: Petrol characteristics

2.3.2 Request 2: Fuel flow rate

A `Matlab` vector was created containing the fuel flow rate $\dot{m}_i$ at a certain time interval i, calculated as the average between the consumed mass of fuel between two time instants, shown in (Fig. 18).

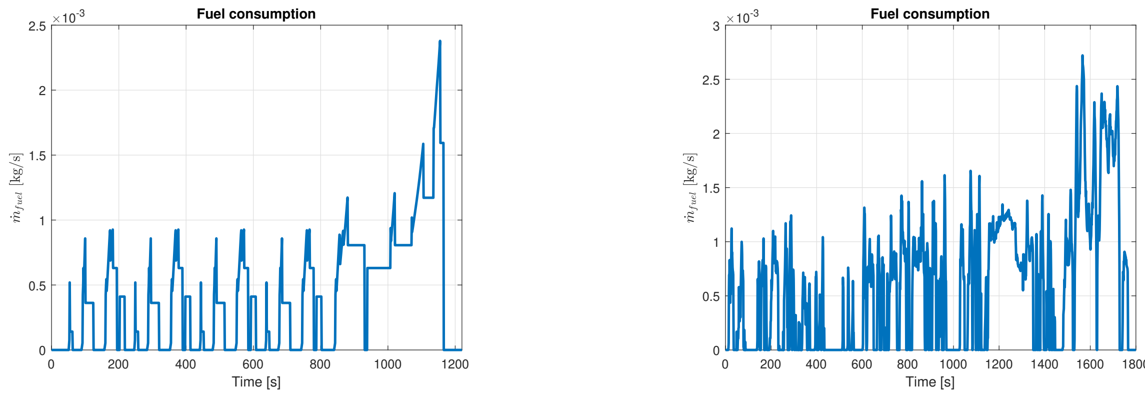
$$\dot{m}_{(i)} = \frac{m_{fuel(i+1)} + m_{fuel(i)}}{2} \cdot \frac{1}{\Delta t_{(i)}} \tag{2.6}$$

2.3.3 Request 3: Total fuel consumption

To compute the total mass of consumed fuel a discrete sum was made in `Matlab`, using a `for` cycle, as described in (Eq. 2.7). Multiplying the fuel flow rate at a given instant with the time interval itself, where N is the number of time intervals. The result is collected in (Tab. 11)

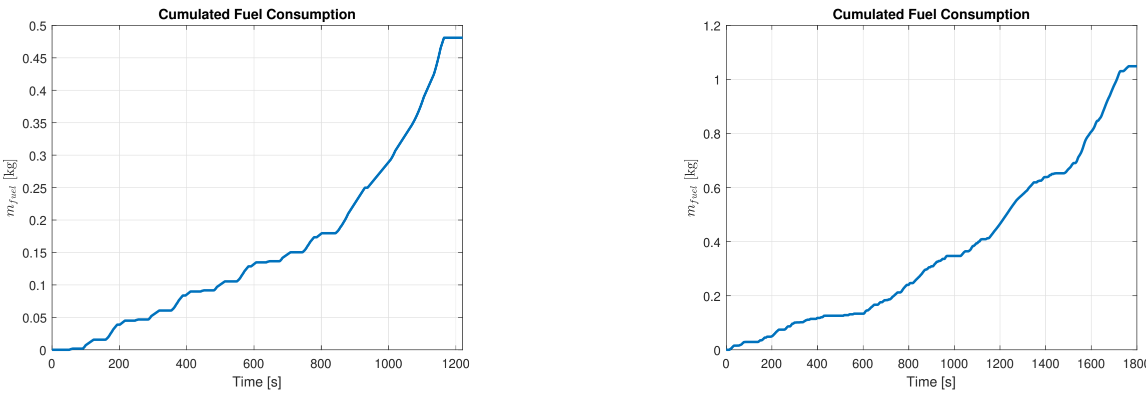
$$m_{tot} = \sum_{i=1}^{N} \dot{m}_{(i)} \Delta t_{(i)} \tag{2.7}$$

2.3.4 Request 4: Fuel flow rate in time



(a) NEDC Cycle (b) WLTP Cycle

Figure 18: Fuel rate $\dot{m}$ along time



(a) NEDC Cycle (b) WLTP Cycle

Figure 19: Cumulated fuel consumption m

2.3.5 Request 5: Tabulated results

NEDC	
Distance	$11.0222\,km$
$\dot{m}_{avg}$	$3.9423 \cdot 10^{-4} kg/s$
m_{tot}	$0.4810\,kg$
L_{tot}	$0.6499\,L$
Declared	$4.4\,L/100km$
Measured	$7.5\,L/100km$
Computed	$5.896\,L/100km$

WLTP	
Distance	$23.2663\,km$
$\dot{m}_{avg}$	$5.827 \cdot 10^{-4} kg/s$
m_{tot}	$1.0489\,kg$
L_{tot}	$1.4175\,L$
Declared	$5.4\,L/100km$
Measured	$7.5\,L/100km$
Computed	$6.092\,L/100km$

Table 11: Consumption results

2.4 Conclusions

The results obtained, even if obtained with some simplifying assumptions, prove to be coherent with the data provided by the manufacturer and by measurements. The declared consumption from the manufacturer is lower than the computed one in both cycles. The reason to that is probably to be found in the conditions which the manufacturer adopted to carried out its calculations: since the aim is to have a car which consumes as little as possible, probably the calculations were made in the most optimal condition possible, leading to the lowest consumption level possible for this vehicle. One of the most influencing parameters which could be unknown is tyres, as rolling resistance has a major role and the vehicle can mount many different tyres, thus probably the manufacturer chose the most efficient tyre possible.
The consumption measuring instead was actually performed in standard real life environment, leading to an expected higher consumptions, due to the non-ideal conditions in which the vehicle could be driven.

Other simplifications that could interfere with the results are:

- Clutch transients: when changing gear, especially in a manual gearbox, the clutch is opened and closed, with a transition period in which a small slip is experienced. This behaviour is not modelled and perfect gear shift are assumed. Moreover, the clutch intervention when accelerating from $n < 1000\ rpm$ was neglected in our analysis.
- Engine efficiency: to extrapolate the values from (Fig. 14) a rough extraction was made by hand to create a lookup table, subsequently the values were interpolated to fit all the combinations of rotational speed and specific work. The resulting values are not completely wrong, but surely a more precise lookup table is needed to have consumption values closer to reality.
- Resistant force: in the computation of the resistance force that the vehicle need to overcome when running, some simplifications were made and some contributions were not considered. For example, all the rotary inertia of the wheels or of the engine mechanisms were not considered in the calculations, as they can be considered as minor contributions. This simplification can create some little discrepancies with the declared and measured values.

3 Assignment 3: Comfort

3.1 Abstract

Aim of the assignment was to analyse the different behaviours that can be observed between cars, that share the same market target, but that were sold in different time periods.
In particular, we were asked to perform an analysis on three Alfa Romeo models:

- Giulietta - 1955/1966
- Alfetta - 1972/1984
- 159 - 2005/2011

The analysis is based on the quarter vehicle car model (Fig. 20), through which different indices that measure the performance of the cars are computed.

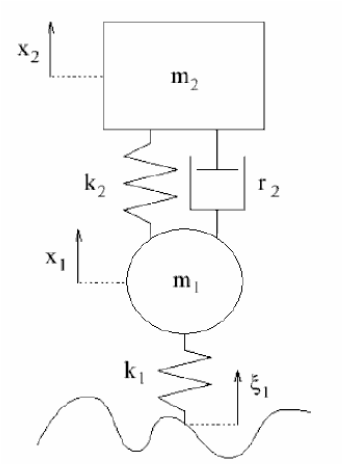


Figure 20: Quarter vehicle model

In the final part, some considerations on the different approaches in the suspension design, between old and new cars, are presented.

3.2 Introduction

To perform the computations, basic cars data are required and summarised in (Tab. 12)

	Giulietta	Alfetta	159
m_1 (kg)	25	30	33
m_2 (kg)	200	270	350
k_1 ($\frac{N}{m}$)	100000	150000	270000
k_2 ($\frac{N}{m}$)	10000	20000	28000
r_2 ($\frac{Ns}{m}$)	1500	2500	5200

Table 12: Car's data

The following indices have been calculated:

- $\sigma_{\ddot{x}_2} = A \cdot \sqrt{\frac{m_1+m_2}{{m_2}^2 r_2}{k_2}^2 + \frac{k_1 r_2}{{m_2}^2}}$
- $\sigma_{\ddot{F}_z} = A \cdot \sqrt{\frac{(m_1+m_2)^3}{{m_2}^2 r_2}{k_2}^2 - 2\frac{m_1 k_1 (m_1+m_2)}{m_2 r_2} + \frac{k_1 r_2 (m_1+m_2)^2}{{m_2}^2} + \frac{{k_1}^2 m_1}{{r_2}^2}}$
- $\sigma_{x_2-x_1} = A \cdot \sqrt{\frac{m_1+m_2}{r_2}}$

With $A = \sqrt{\frac{A_b v}{2}}$

A_b is an index that describe how much is the amplitude of the road displacement. For a value of $A_b = 1.4 \cdot 10^{-5}\ m$ we have a 'Normal road', while for $A_b = 1 \cdot 10^{-4}\ m$ the road can be described as 'Rough'.

These indices are the Discomfort, Road Holding and Working Space, and they need to be as low as possible. Moreover, the frequency response function of the sprung mass vertical displacement has been calculated for the three cars.

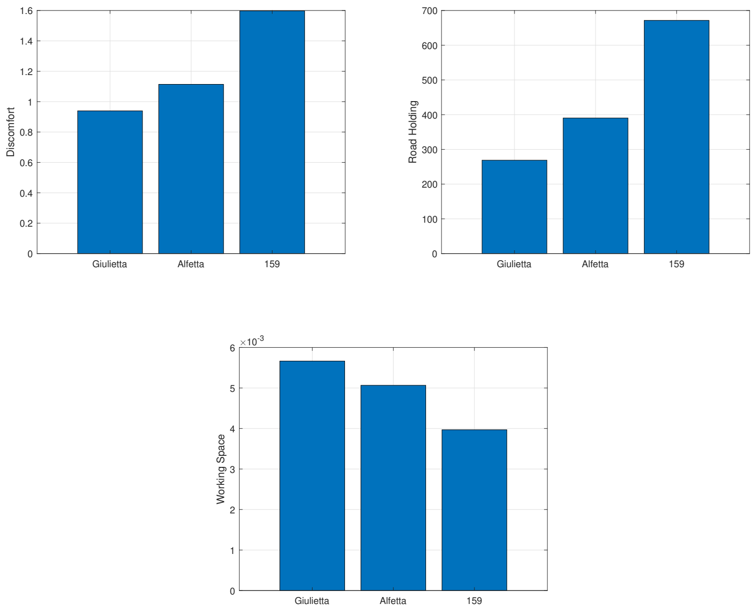


Figure 21: Car indexes for $A_b = 1.4 \cdot 10^{-5}\ m$

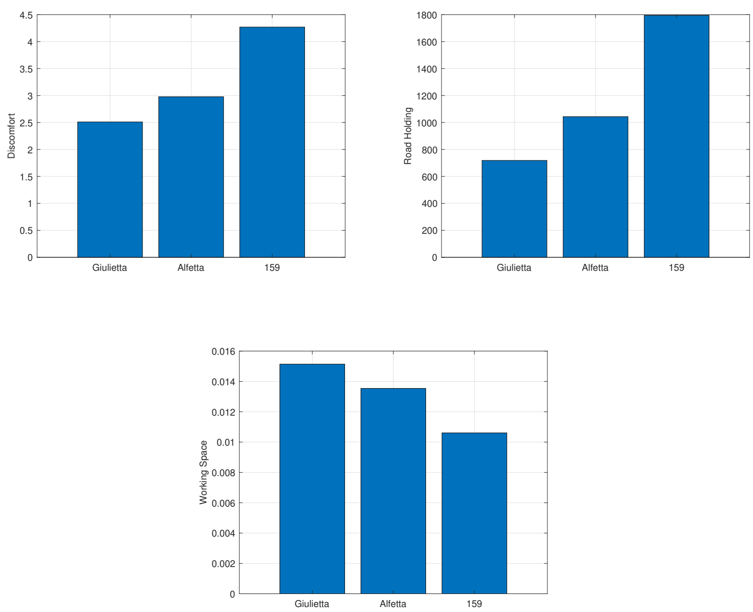


Figure 22: Car indexes for $A_b = 1 \cdot 10^{-4}\ m$

As can be observed, the Giulietta is the most comfortable car and the one whose shows the lowest amount of Road Holding. In general the trend for Discomfort is worsening over the years, which can be explained thinking to the evolution of the road in which the vehicle is expected to run, which instead got better. The transfer function between the sprung mass displacement and the road displacement has been calculated for the three cars.

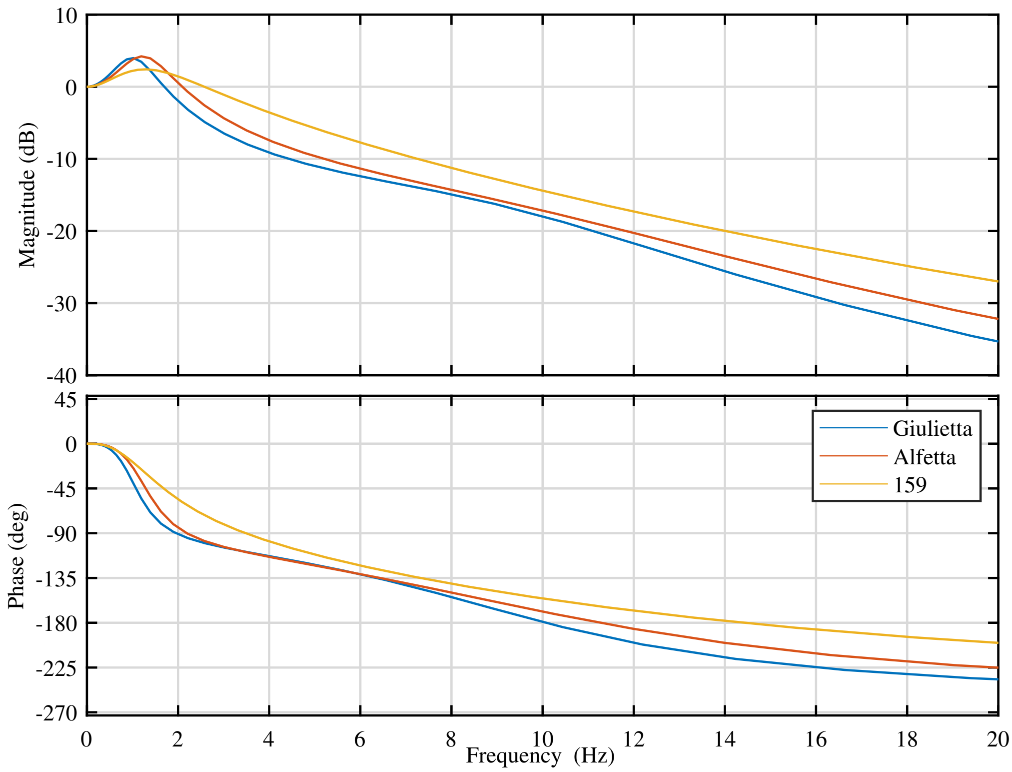


Figure 23: Displacement Transfer functions

The 159 performs better in a limited range of low frequencies, but in the range above 2 Hz is worse. In the vertical direction, the human body can easily resist excitations of around 1 Hz, but between 10 to 20 Hz excitations can be felt as more uncomfortable due to the human body resonances.

3.3 Influence of parameters on the Giulietta

In the following plots, (Fig. 24) and (Fig. 25), the influence of the parameters on Discomfort and Road Holding of the Giulietta are presented.
As can be observed, the value for k_2 is chosen in order to minimize the Road Holding.

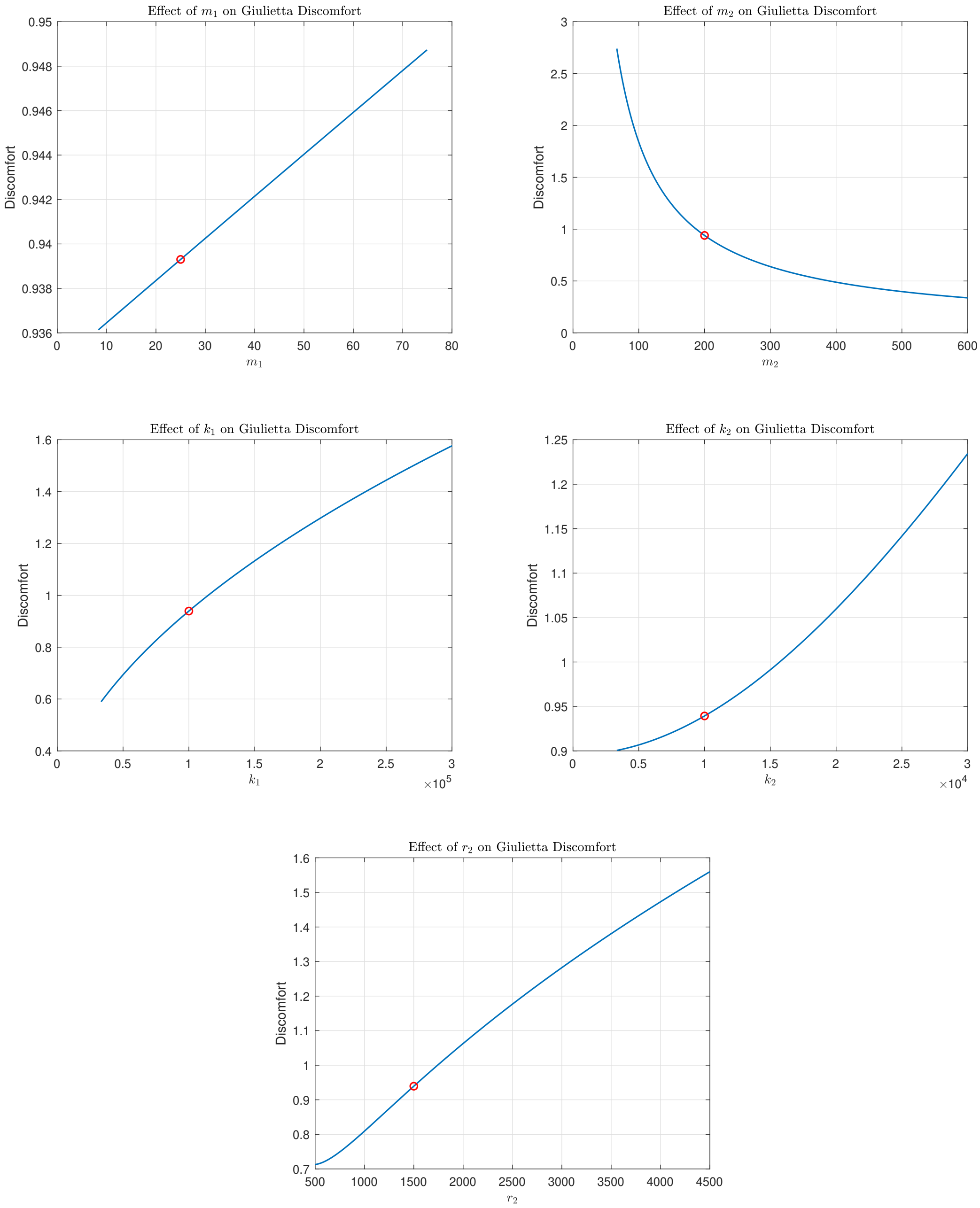


Figure 24: Parameters vs Discomfort

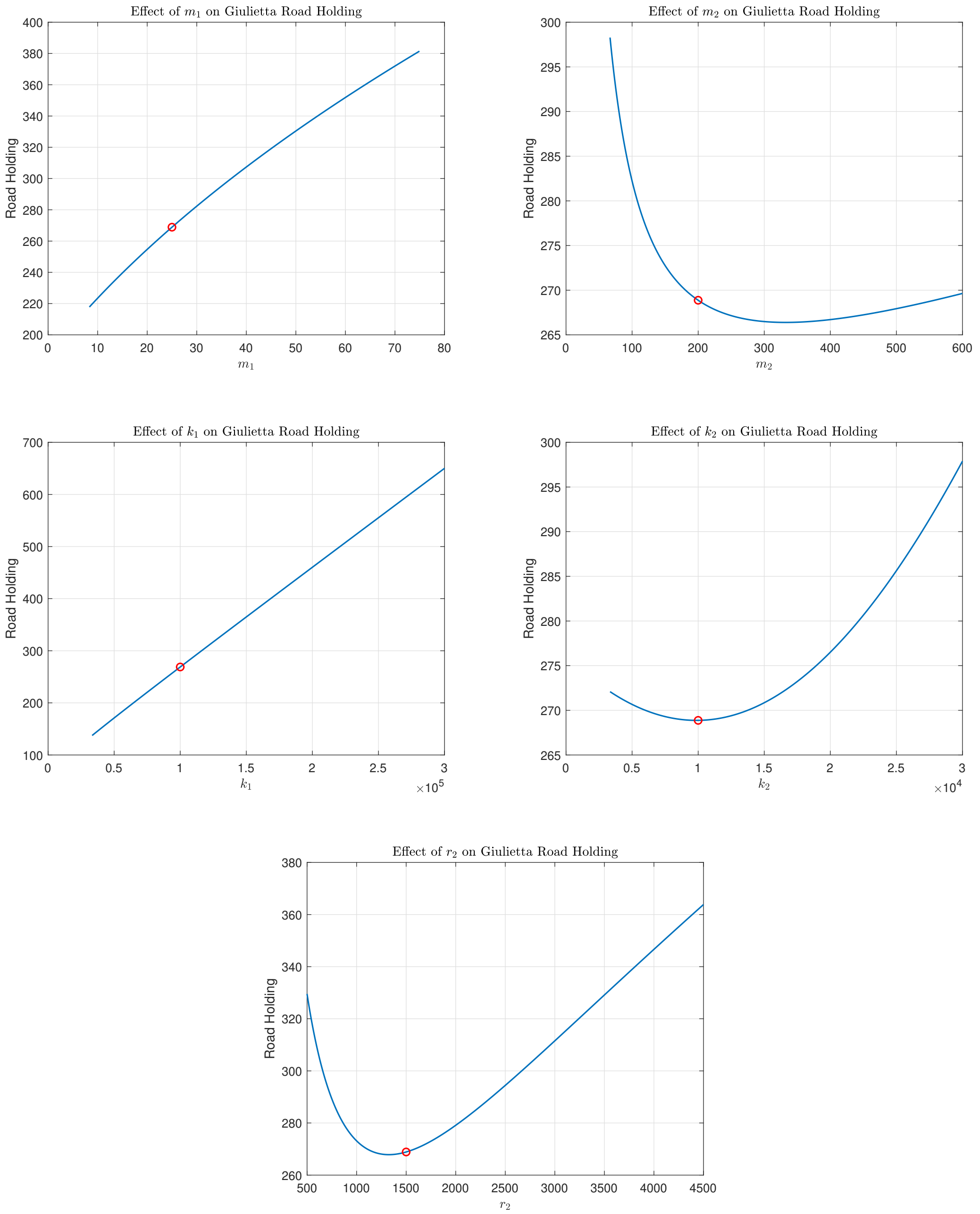


Figure 25: Parameters vs Road Holding

3.4 Conclusion

We computed, through the quarter vehicle model, the Discomfort, Road Holding and Working Space for the cars, each one representing different epochs in the car history.
The Discomfort of cars has rise up during the years, which can be explained due to the fact that roads got better and better in terms of irregularities, while in the 60s they were more bumpy.
The transfer function of the sprung mass vertical displacement has been computed for the three models in order to investigate the difference between them, underlying a better Discomfort at high frequency for the older models.
After that, the influence of the various parameters of the model on the Giulietta car has been investigated. The selected stiffness of the suspension is the one which minimizes the Road Holding. A decrease in damping corresponds to an improvement in the Road Holding and Discomfort, but as a drawback the working space will increase.

4 Assignment 4: MacPherson LCA design

4.1 Abstract

The suspension of a road vehicle is a vital component in ensuring its best behaviour in terms of comfort and handling. In this assignment, the design of a MacPherson road vehicle suspension is treated, with a particular focus on the static assessment of the lower control arm.
The hard-points of the suspension were first calculated through a Matlab Sim-mechanics model and after that, a first design of the control arm has been developed considering the available room and kinematic requirements, after having performed a pre-dimensioning procedure. The control arm has then been optimized through the use of the *Altair Inspire* software, aiming to reduce its weight. As a last step, the lower control arm has been tested through a Finite Elements Method, by exploiting the *Dassault Systèmes Abaqus* software, to perform a static assessment on the final design with the given loads. The missing components of the suspension have been designed to produce a full scale drawing of the assembly, while a complete production drawing has been elaborated for the control arm only.

4.2 Introduction

The vehicle suspension should be designed to enhance comfort and ensure proper road holding capabilities. A large variety of road vehicles adopt the MacPherson suspension type, characterized by the presence of the lower control arm only and a spring-damper element in place of the upper control arm.
This kind of suspension design is really popular in the automotive industry because guarantees adequate performance at a relatively low cost and low space requirements, with respect to other solutions, such as the double wishbone suspension.

The considered suspension is the one of a front wheel of a front wheel drive vehicle.

The choice of the hard-points position has been performed through a 3D kinematic model developed in Matlab Sim-mechanics, with the objective of matching some target values of camber angle and toe angle during the wheel vertical travel. This choice has been discussed in detail in (Sec. 4.3).

Then an estimation of the suspension loads has been done and discussed in (Sec. 4.4), considering an extremely demanding and unlikely load condition, to have a high safety margin in the control arm dimensioning process.

The pre-design of the lower control arm is discussed in (Sec. 4.5), while its weight reduction and FEM analysis are discussed in (Sec. 4.6).

Finally, the drawings of the lower arm and the assembly have been developed through the use of the *Dassault Systèmes Catia V5* software.

4.3 Suspension's joints coordinates selection

Assuming that a dynamic study on the vehicle's performance was already carried out, it was requested to satisfy some kinematics objectives by choosing the coordinates of some particular suspension joints.

4.3.1 Objective and constraints

The kinematic objectives to be satisfied were regarding the camber angle and steering angle, which had to reach a determined value at different values of wheel travel. In particular, the angles were studied at the nominal position of the spring-damper assembly, at a full compression of $80\,mm$ and at full extension of $95\,mm$, the objectives are reported in (Tab. 13).

	Nominal	Compression	Extension
Camber	$-1°$	$-0°26'30''$	$-0°28'24''$
Steering	$-0°03'17''$	$-0°46'22''$	$-0°26'10''$

Table 13: Kinematic objectives

To reach these kinematic objectives the coordinates of some particular points were changed, respecting room constraints reported in (Tab. 14). The vehicle reference system is centred in a point inside the mean plane of the vehicle, with x axis pointing rearwards along the longitudinal direction, z is the vertical axis pointing upwards and y is created to make a right-handed reference system.
The studied points were:

- Point B: joint between wheel hub carrier and lower arm
- Point C_1: joint between chassis and lower arm
- Point G: steering joint on the chassis

	$y\,[mm]$	$z\,[mm]$
B	$715:735$	$-50:-70$
C_1	$390:410$	$-45:-65$
G	$300:320$	86

Table 14: Room constrains

4.3.2 Optimisation problem

To reach the requested objectives, a optimization problem was set up, with the aim of reaching a good solution in a structured and precise way, using a limited amount of time and resources. It was decided to use a uniform grid approach, even if slow and not precise, due to its robustness and certainty of result.
The problem was set up as follows:

- Design variables: It was chosen to use as design variables all the varying joint coordinates pointed out in (Tab. 14). A `linspace` command was used in `Matlab` to create a vector of a certain number of elements for each design variable, using as limits (Tab. 14), subsequently they were combined and then inserted into the function to compute the requested angles.
- Angles computation: Using a provided `SimMechanics` model, steering and camber angle, at different compression lengths were computed, for each combination of design variable.

- Objective functions: After extracting the requested angles the objective functions for the optimisation method were computed. It was decided to use the mean square error of the computed angle respect to the target one, at each compression values, more precisely:

$$f_{camber} = \frac{(c_nom - c_{nom,t})^2 + (c_{compr} - c_{compr,t})^2 + (c_{exts} - c_{exts,t})^2}{3} \tag{4.1}$$

$$f_{steering} = \frac{(s_nom - s_{nom,t})^2 + (s_{compr} - s_{compr,t})^2 + (s_{exts} - s_{exts,t})^2}{3} \tag{4.2}$$

- Pareto front extraction: The objective functions were evaluated and an objective function domain was created. Inside the OF domain an optimization algorithm, containing the definition of a Pareto optimal condition, was run and a Pareto front extracted.

Due to the high number of possible combinations and long computational time, the simulation could only be run initially with a very coarse DV domain, leading to the results in (Fig. 26). Here it was evident the presence of at least one DV generating the clustering of the solutions in similar displaced areas of the domain, with a replicated trend. Even if the Pareto front was the optimal solution for this set of DV, the error values for each objective function were still very high, and the results were not acceptable.

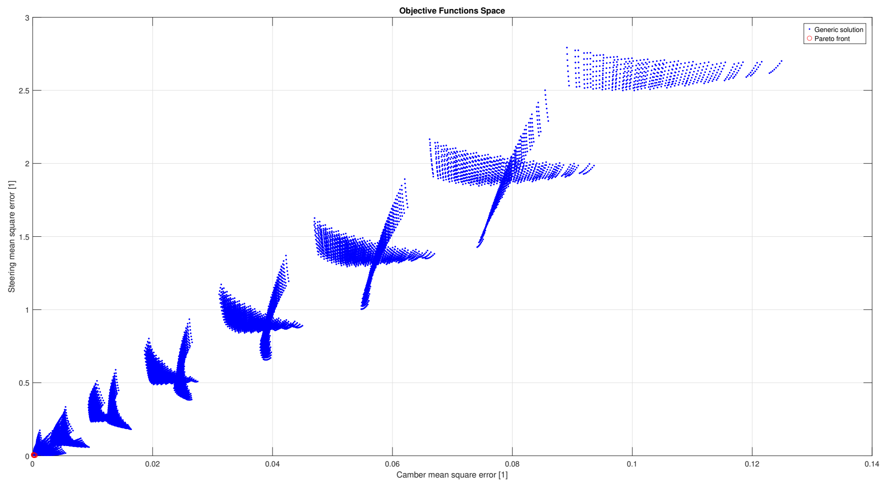


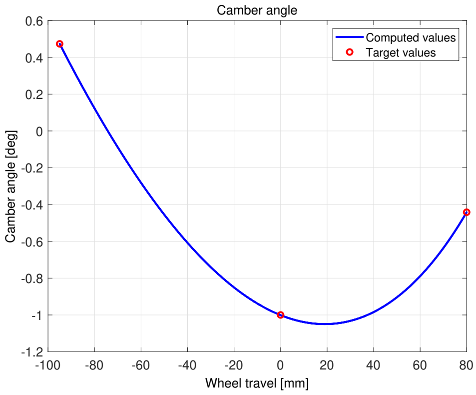
Figure 26: Coarse OF domain

Moreover, by analysing the DVs of the few Pareto solutions found, it could be noticed that they had three constant DVs, corresponding to one of their boundaries, thus could be excluded by the combinations by keeping them constant. Subsequent trials using this approach provided insufficient results again. This characteristic of the OF domain was found to be dependent on the discretisation of the DVs, which was very coarse.
Therefore other two simulations were launched, with a finer discretisation of the DVs. To avoid the running too computationally time demanding simulations, the number of elements in the `linspace` was kept low, but the boundary of each DV was reduced, centering it around the DVs that made the Pareto front of the first simulation in (Fig. 26). In this way it was possible to obtain a finer discretisation of the DVs domain, closer to the actual Pareto front.

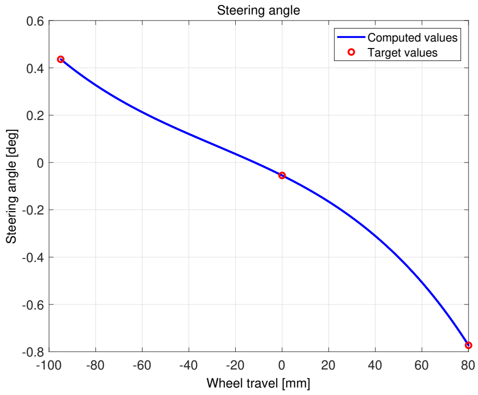
This method proved to be successful, and within the Pareto Optimal solutions, the following was chosen (Tab. 15), which provided these results for steering and camber angle (Fig. 27). Here it is clear that each objective was satisfied.

B_y	B_z	$C1,y$	$C1,z$	G
$726\,mm$	$-56\,mm$	$406.4\,mm$	$-50\,mm$	$317.2\,mm$

Table 15: Chosen joint coordinates



(a) Camber angle



(b) Steering angle

Figure 27: Optimisation results

The final coordinates necessary to build the suspension are reported in (Tab. 16)

	$x\,[mm]$	$y\,[mm]$	$z\,[mm]$
A	31.8	593	604
B	-9.1	726	-56
C_1	-10	406.4	-50
C_2	311	406.4	-45.4
D	-1.6	773.5	-226.8
E	-1.6	768.4	65.1
F	136.9	682.7	88.5
G	170	317.2	86
O	0	0	0

Table 16: Chosen joints coordinates

4.4 Internal forces computation

For the computation of the loads acting on the suspension system, three different contributions have been taken into account, corresponding to a vertical force (Z), a lateral one (Y) and a third longitudinal load (X). The three components are computed in (Eq. 4.3).

$$\begin{cases} Z = KF_{z,st} \\ Y = \mu_y Z \\ X = \mu_x Z \end{cases} \tag{4.3}$$

Where:

$$\begin{cases} F_{z,st} = gm_{s+u} = 9.81\ m/s^2 \cdot 390\ kg \simeq 3825\ N \\ K = 2.5 \\ \mu_y = 0.7 \\ \mu_x = 1.2 \end{cases}$$

Z is generated by a bump hit by the wheel and the factor K represents the dynamic effects on the load.
Y is generated during a turn and represents the cornering forces.
X can in general describe both accelerating or braking depending on the considered direction.

Note that our case study is an extreme one, in which a bump is hit during a turn, while braking/accelerating. All the resulting loads and stresses are thus sufficiently large to suppose a safe approach which is indeed exploited just for pre-dimensioning purposes. A more precise analysis should be performed for greater accuracy of the stresses (also with fatigue analysis and other kinds of assessment).

The main idea in computing the loads is that the arms of the suspension system are considered to be rods, meaning that they are capable of transmitting only axial forces. For this reason, the final goal is the computation of the in-plane components of the loads with respect to the control arm to be pre-dimensioned (represented by points B, C_1 and C_2).

The loads to be computed are represented in (Fig. 28).

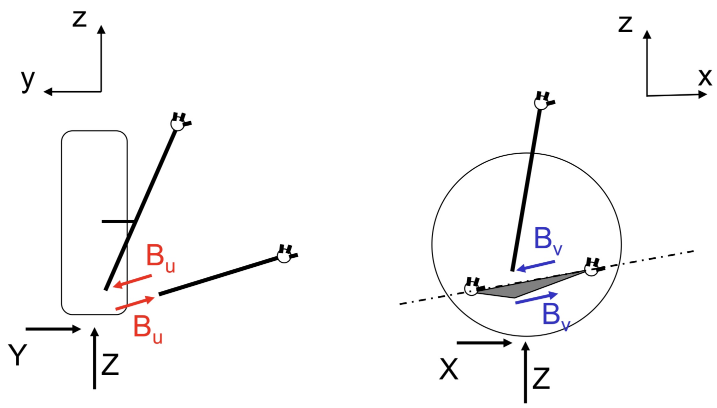


Figure 28: In-plane loads

Starting from the calculation of B_u, by exploiting the points summarized in (Tab. 16) and considering the scheme of (Fig. 29), the load is computed through a moment equilibrium about point A, as shown in (Eq. 4.4).

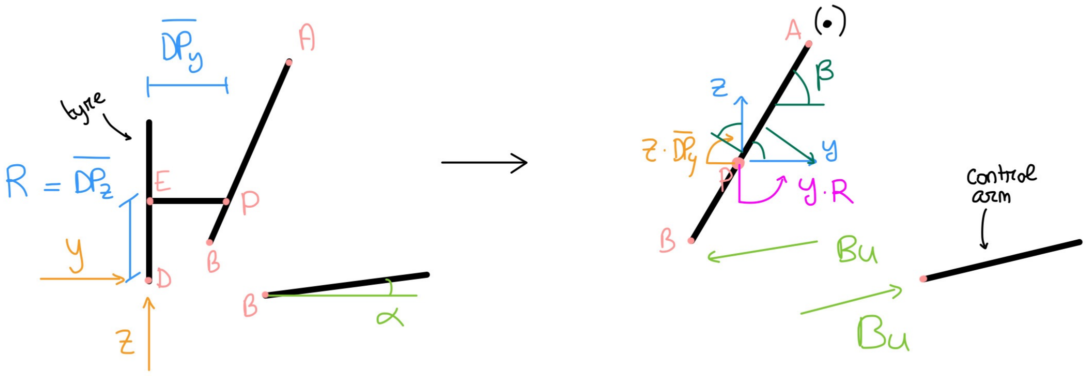


Figure 29: Structural scheme - YZ plane

$$Y \cdot R - Z \cdot \overline{DP_y} + (Y \cdot \sin\beta) \cdot \overline{AP_{yz}} - (Z \cdot \cos\beta) \cdot \overline{AP_{yz}} - B_u \sin(\beta - \alpha) \cdot \overline{AB_{yz}} = 0 \tag{4.4}$$

Where

$$\begin{cases} R = \overline{DP_z} = \overline{DE_z} = 291.9\ mm \\ \overline{DP_y} = \overline{AD_y} - \overline{AP_y} = 156.8\ mm \\ \overline{AP_y} = \overline{AP_z} \cdot \tan(90 - \beta) = 23.65\ mm \\ \overline{AP_z} = \overline{AE_z} = 538.9\ mm \\ \overline{AP_{yz}} = \sqrt{\overline{AP_y}^2 + \overline{AP_z}^2} = 539.4\ mm \\ \overline{AB_{yz}} = \sqrt{\overline{AB_y}^2 + \overline{AB_z}^2} = 673.3\ mm \\ \beta = \arctan\frac{\overline{AB_z}}{\overline{AB_y}} = 78.6\ deg \\ \alpha = \arctan\frac{\overline{BC_{1z}}}{\overline{BC_{1y}}} = 1.0755\ deg \end{cases}$$

Note that, being this a first tentative computation, the point C1 and C2 are considered to be at the same position in the YZ plane, for the computation of the angle α. Finally:

$$B_u = \frac{Y \cdot R - Z \cdot \overline{DP_y} + (Y \cdot \sin\beta - Z \cdot \cos\beta) \cdot \overline{AP_{yz}}}{\sin(\beta - \alpha) \cdot \overline{AB_{yz}}} = 5835\ N \tag{4.5}$$

With the same reasoning, the force B_v is also computed.

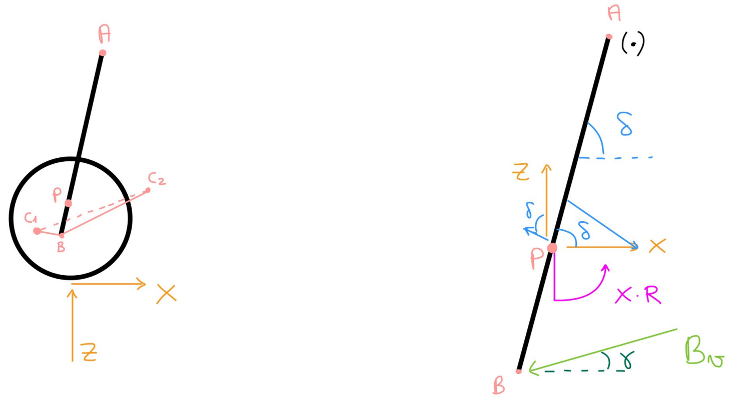


Figure 30: Structural scheme - XZ plane

Note that the braking condition is considered, being it the worst possible case due to the generation of a moment XR coming from the brake caliper which is connected to the hub carrier, as a reaction to the braking action.

Also in this case, a moment equilibrium is performed about point A.

$$X \cdot R + (X \cdot \sin\delta) \cdot \overline{AP_{xz}} - (Z \cdot \cos\delta) \cdot \overline{AP_{xz}} - B_v \cdot \sin(\delta - \gamma) \cdot \overline{AB_{xz}} = 0 \tag{4.6}$$

Where

$$\begin{cases} \overline{AP_{xz}} = \sqrt{\overline{AP_x}^2 + \overline{AP_z}^2} = 593.9\ mm \\ \overline{AP_x} = \overline{AP_z} \cdot \tan(90 - \delta) = 33.35\ mm \\ \overline{AB_{xz}} = \sqrt{\overline{AB_x}^2 + \overline{AB_z}^2} = 661.3\ mm \\ \delta = \arctan\frac{\overline{AB_z}}{\overline{AB_x}} = 86.5\ deg \\ \gamma = \arctan\frac{\overline{C_1C_{2z}}}{\overline{C_1C_{2x}}} \end{cases}$$

Which, rearranging (Eq. 4.6), results in:

$$B_v = \frac{X \cdot R + (X \cdot \sin\delta - Z \cdot \cos\delta) \cdot \overline{AP_{xz}}}{\sin(\delta - \gamma) \cdot \overline{AB_{xz}}} = 13978\ N \tag{4.7}$$

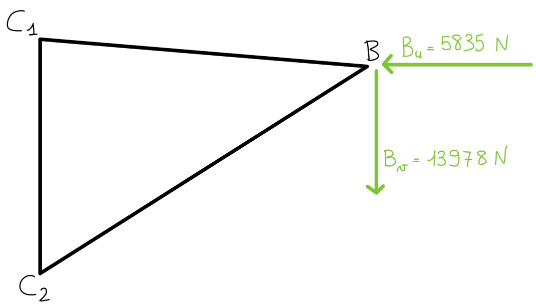


Figure 31: Final forces - Control Arm plane

Now, the focus turns to the computation of the reaction forces at the points C_1 and C_2. At these locations, two different bushings are positioned, each one with proper stiffnesses. In particular, the two radial stiffnesses (the ones oriented as the force B_u) share the same value, while a correct ratio between the axial ones needs to be selected. The distribution of the reaction forces depends on this ratio. In principle any ratio could be chosen, but a possibility is to consider one stiffness much larger than the other, in order to approximate the constraints to a hinge (larger stiffness thus smaller compliance) and a roller (smaller stiffness thus larger compliance): for our analysis, this solution has been taken into account, so to obtain a statically determined structure which can be more easily reproduced also in both *Dassault Sistèmes Abaqus* and *Altair Inspire* softwares, as better explained in (Sec. 4.6).

The hinge has been positioned at point C_2, the roller at point C_1, has shown in (Fig. 32)

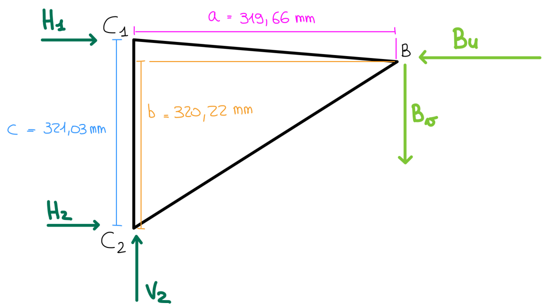


Figure 32: Reaction forces - Control Arm plane

The forces H_1, H_2 and V_2 are computed through a system of three equations: horizontal forces equilibrium, vertical forces equilibrium, moment equilibrium about point C_2 (Eq. 4.8)

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -c & 0 & 0 \end{bmatrix} \begin{bmatrix} H_1 \\ H_2 \\ V_2 \end{bmatrix} = \begin{bmatrix} B_u \\ B_v \\ B_v a - B_u b \end{bmatrix} \tag{4.8}$$

Which results in:

$$\begin{cases} H_1 = -8097\,N \\ H_2 = 13933\,N \\ V_2 = 13978\,N \end{cases}$$

4.5 Section pre-dimensioning and steering angle check

The first issue for the pre-dimensioning process is to define a first tentative shape of the control arm itself, which needs to ensure the correct functioning of all the required kinematics. In particular, the wheel must be able to perform a 37° steering angle inward and a 32° one outward at the nominal suspension position. The preliminary shape is shown in (Fig. 33).
The steering angles have been verified and shown in (Fig. 34).

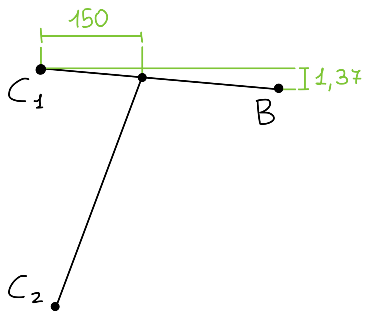
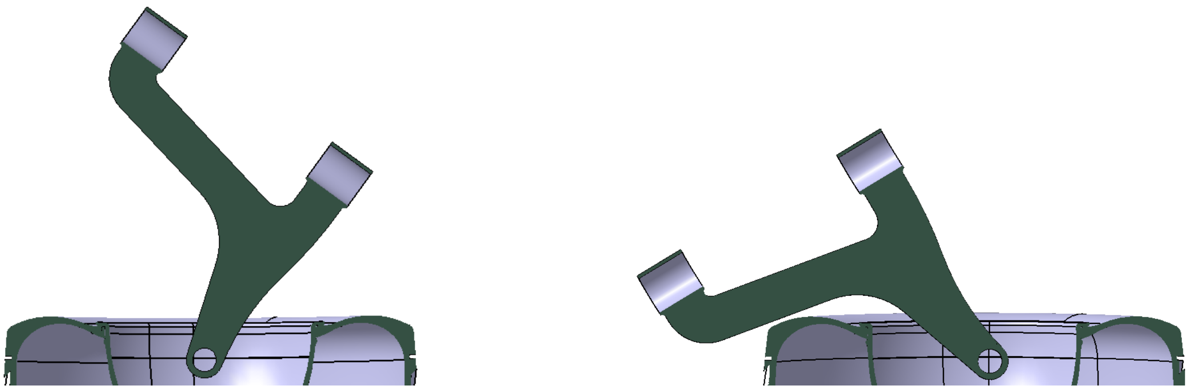


Figure 33: Schematic preliminary shape - Control Arm plane



(a) 37 *deg* inward

(b) 32 *deg* outward

Figure 34: Steering angles check with a preliminary shape of the CA

To simplify the computation (always keeping in mind that we are referring to a first approximation solution), the structure is approximated as shown in (Fig. 35)

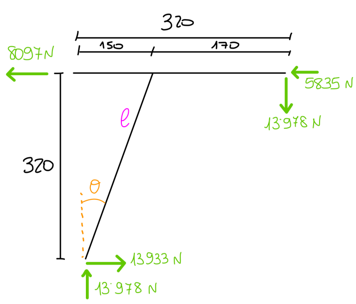


Figure 35: Schematic simplified shape - Control Arm plane

Considering that $l \approx 353.4\ mm$, $\theta \approx 25.11\ °$ and the fact that the most stressed region is located where the two arms starting from C_1 and C_2 meet, the pre-dimensioning is performed with the following steps.

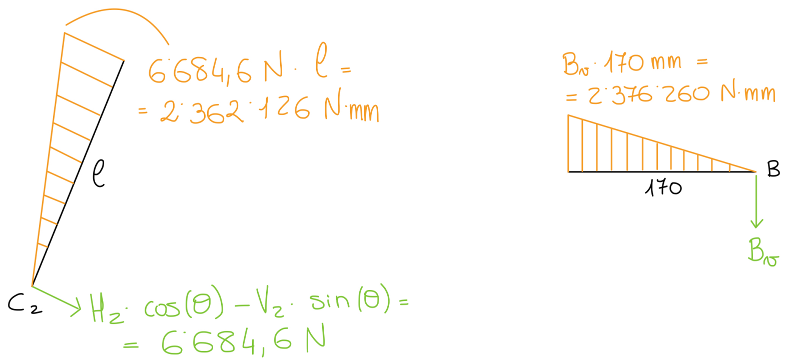


Figure 36: Bending moment diagrams (axial stresses are negligible)

At first, the maximum bending moment is computed:

$$M_b = 13978\ N \cdot 170\ mm = 2376260\ N \cdot mm$$

Then, a rectangular cross-section is considered, where the height is assumed to be one third of the base, as shown in (Fig. 37).

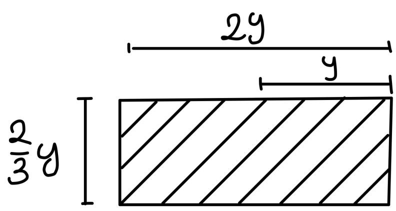


Figure 37: Cross-section

The cross-section moment of inertia is:

$$I = \frac{\frac{2}{3}y(2y)^3}{12} = \frac{4}{9}y^4$$

The maximum stress in the section is:

$$\sigma_{max} = \frac{M_b \cdot y}{I} = \frac{9M}{4y^3}$$

Considering 1.5 as safety coefficient (even if the considered load are already large) and selecting the *Aluminium A357* T6 as material (yielding stress $\sigma_y = 270\,MPa$):

$$\sigma_{max} \leq \frac{\sigma_y}{1.5}$$

$$y \geq \sqrt[3]{\frac{9 \cdot 1.5 \cdot M_b}{4 \cdot \sigma_y}} \simeq 30\,mm$$

Finally, the control arm cross-section results to be 60 mm wide and 20 mm large, as drawn in (Fig. 38), resulting in the geometry in (Fig. 39).

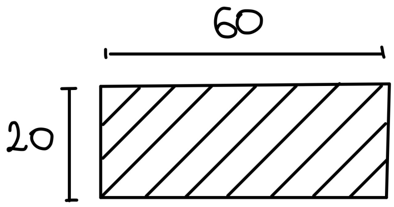


Figure 38: Final cross-section

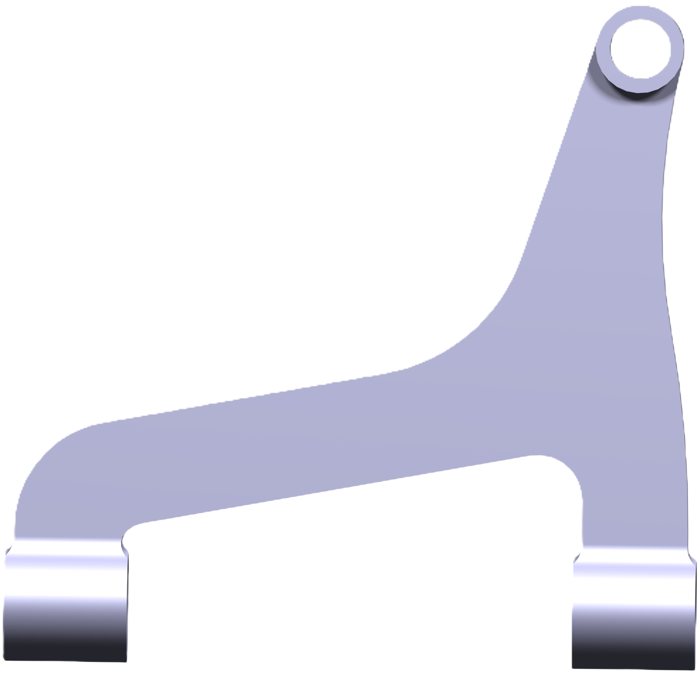


Figure 39: LCA predimensioning render

4.6 FEM analysis and weight reduction

After the pre-dimensioning phase, a first FEM analysis using *Dassault Sistèmes Abaqus* was carried out in order to verify that the stresses occurring in the LCA are below the yield stress limit of the material. After a satisfactory pre-dimensioning section was found, a structural optimisation analysis was carried out using *Altair Inspire*, with the aim of identifying regions where material could be removed, reducing the overall weight without affecting the overall stiffness of the component. In the end, a second, more thorough, FEM analysis was carried out to validate the design choices, using finer mesh and better elements.

4.6.1 Pre-dimensioning check

During the pre-dimensioning check phase, two different whole sections were analysed:

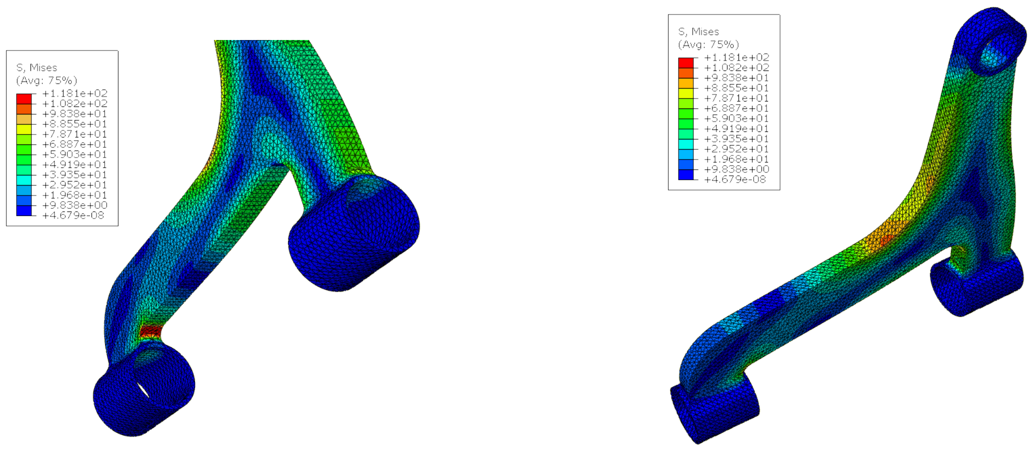
- The original section: $[60 \times 20]\ mm^2$
- A smaller section: $[50 \times 16]\ mm^2$

After analysing the original section it was clear that it was oversized and resulted in very low Von Mises stresses compared to the yield stress of the aluminium A357 T6. Therefore it was decided to reduce straight away the overall section of the LCA, still keeping the same ratio between length and width equal to 3, resulting in the second, smaller, section. Here the maximum Von Mises stress increased, lowering the safety factor η down to 1.44.

	σ_{VM}^{max}	σ_{yield}	η
$[60 \times 20]$	$118\ MPa$	$270\ MPa$	2.29
$[50 \times 16]$	$188\ MPa$	$270\ MPa$	1.44

Table 17: Pre-dimensioning results

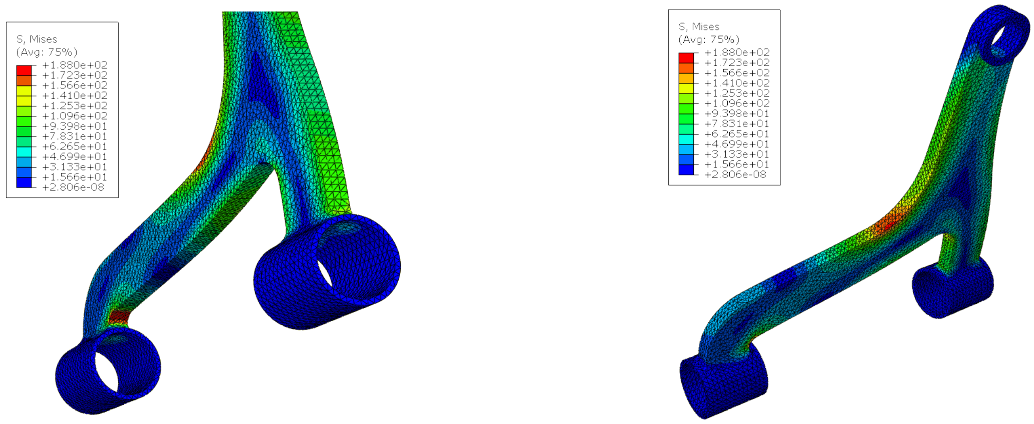
As expected, the highest stresses were localised in two main regions: the region near the point C_2, where the material has a very tight radius just close to the housing of the bushing, and in the curve where the piece of the arm coming from C_2 connects with the part that spans from B to C_1.



(a) Housing connection close up

(b) Bend close up

Figure 40: Close up of highly loaded regions for the 60 mm section



(a) Housing connection close up

(b) Bend close up

Figure 41: Close up of highly loaded regions for the 50 mm section

4.6.2 Topology optimization with Altair Inspire

A topology optimization with the aim to reduce the LCA weight was run using *Altair Inspire*. The optimization was set up to maximise the stiffness of the LCA while reducing the mass of at least 50%, by removing material where not needed. The boundary conditions applied allowed the LCA to rotate along the bushing's axis, with the previously mentioned roller-hinge configuration.

The results shown in (Fig. 42) suggested that the central part of the LCA was lightly loaded, therefore the removal of material in that region would not affect considerably the overall stiffness. Other regions could be removed but, for safety reasons, it was decided not to excessively reduce the overall stiffness in an unnecessary way. Moreover, different runs inside *Altair Inspire* at different settings always suggested to remove the central part, independently of the material removal value. In addition, it was clear also from the *Abaqus* results that very low stresses characterised that region.
The LCA geometry was updated in *CatiaV5*, removing the aforementioned region.
The optimization resulted in a mass reduction from 1.52 kg to 1.43 kg, corresponding to a total removal of almost 0.1 kg or around 6 %, very important for handling performances.

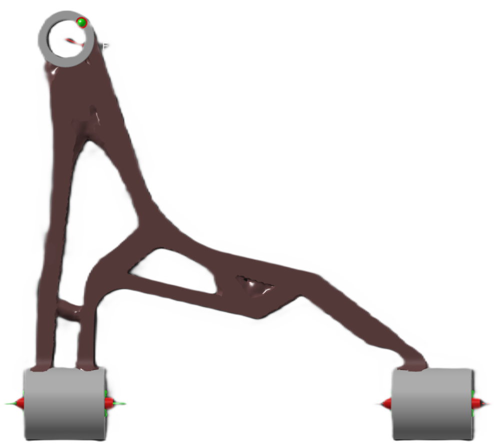


Figure 42: Optimization result using *Altair Inspire*

4.6.3 Final FEM analysis

After removing the unnecessary material another FEM analysis was carried out, which suggested that further modifications could be made to further reduce the material needed. It was decided to slightly change the LCA section shape, passing from the rectangular section of (Fig. 38) to an H-shaped one: after many iterations the chosen section was the one reported in (Fig. 43).

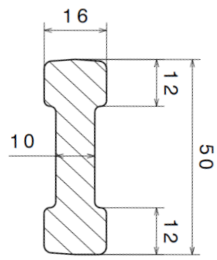


Figure 43: H-shaped section

The FEM analysis carried out with *Dassault Sistèmes Abaqus* with the following setup:

- Reference system: Centred in C_1 with the x axis passing through C_2, y axis towards B and the z axis perpendicular to them.
- Material: Al A357 T6 with $\sigma_y = 270\,MPa$ and $\nu = 0.333$, mechanical elastic behaviour.
- Step: Static, linear perturbation
- Boundary conditions: Isostatic system with a realistic behaviour allowing the rotation around the bushing axis, the boundary condition was rigidly coupled to the entire internal cylinder holding the bushing or the ball joint.

Point	BC	Dof
B	Support	All except U_z
C_1	Roller	U_x, UR_x
C_2	Hinge	UR_x

Table 18: Boundary conditions

- External loads: The loads B_u and B_v computed in (Sec. 4.4), were applied to the B point, along the positive x and negative y direction respectively.
- Mesh: C3D20 twenty node tetrahedric element, with full integration, seed dimension of 5, with some local refinements in the most loaded and critical regions, such as the central bend and the connection of the bushing's housing.

The FEM analysis shows that the central part of the LCA, where the material was removed, was indeed lightly loaded, and the presence of a hole did not induced any stress-related issue in the region. Moreover, the LCA has particularly high Von Mises stresses along the bend (Fig. 45b) and in the corner where the LCA connects to the bushing representing the hinge connection (Fig. 45a). In the bend region the high stresses are due to the load B_v, which is particularly large along the x direction, while in the corner region the high concentration is due to the small radius and the presence of both radial and axial reaction forces.

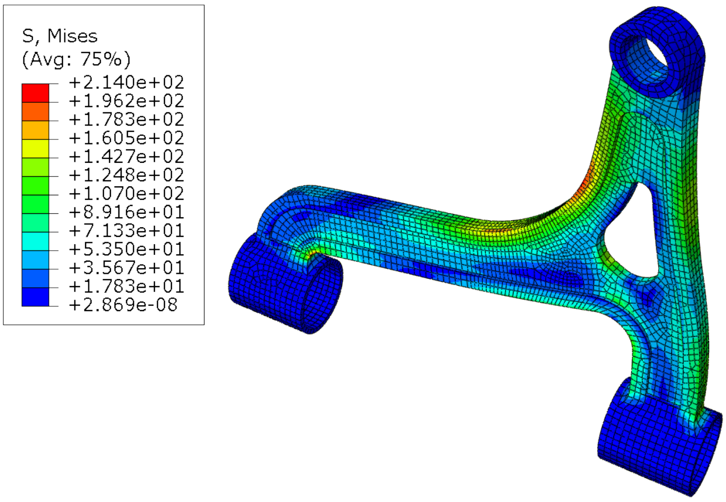
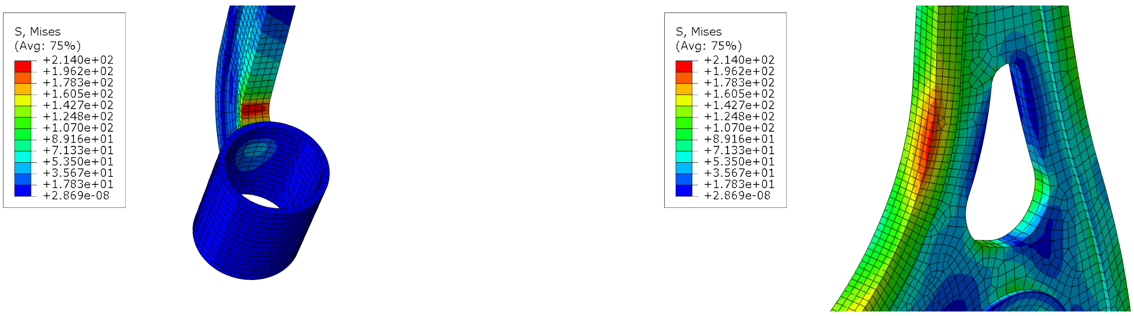


Figure 44: Von Mises stress



(a) Housing connection close up

(b) Bend close up

Figure 45: Close up of highly loaded regions

The safety coefficient was calculated using the maximum Von Mises stress computed by the software, which was equal to $\sigma_{VM}^{max} = 214\,MPa$:

$$\eta = \frac{\sigma_{yield}}{\sigma_{VM}^{max}} = \frac{270\,MPa}{214\,MPa} = 1.26 \tag{4.9}$$

Even if the safety coefficient is relatively low, the LCA results verified. Moreover, the safety coefficient achieved is close to the one used in (Sec. 4.5) to compute the initial section, even after all the modifications made in order to reduce the overall LCA weight. The load scenario applied of braking and hitting a bump while turning is highly unlikely and strongly demanding, therefore even if the safety coefficient is not particularly high, it can be regarded as an acceptable quantity, effectively passing the static load tests.

The final product, with all the needed features such as fillets and draft angles, is shown in (Fig. 46) and (Fig. 47).

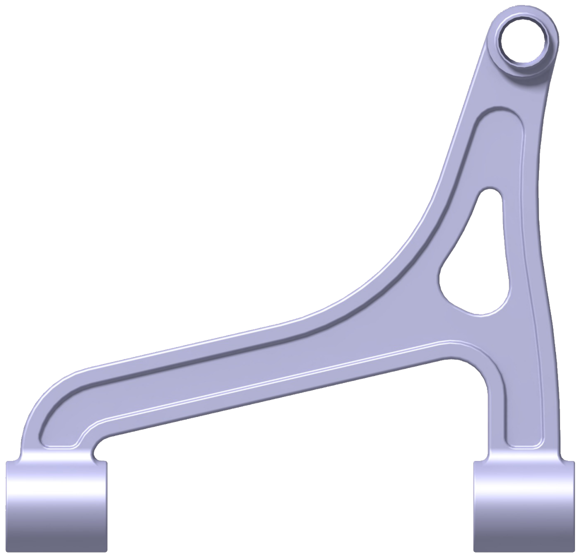


Figure 46: LCA Final render front view

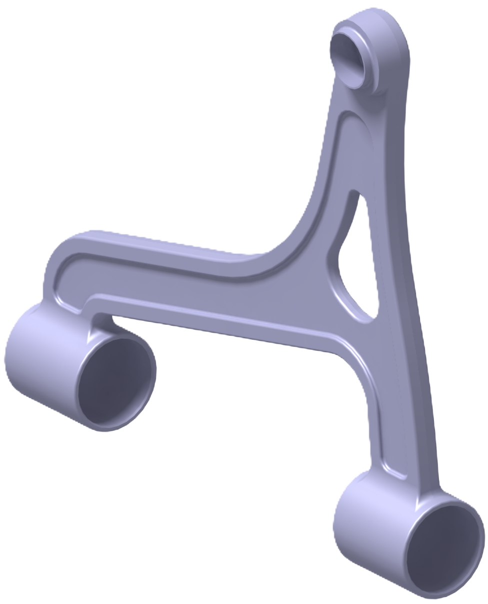


Figure 47: LCA Final render isometric view

4.7 Conclusions

In conclusion, the design and analysis process of the MacPherson lower control arm (LCA) has been carried out with a methodical approach, addressing both kinematic and structural requirements. The project started with the selection of suspension joints' coordinates aimed at ensuring optimal camber and steering angles across different wheel travel conditions. Subsequently it was possible to compute the internal forces to understand the loading conditions, leading to a well-informed pre-dimensioning phase.

The pre-dimensioning process provided a solid foundation for the FEM analysis, where different cross-sectional configurations were tested to ensure structural integrity while minimizing weight. The initial rectangular section, after many iterative simulations, was optimized in a H-shaped section that effectively balanced strength and mass reduction. The topology optimization performed with Altair Inspire further contributed to material reduction in non-critical areas, achieving a significant decrease in overall weight without compromising the mechanical performance.

The final FEM analysis confirmed that the optimized design meets the required performance criteria, with a safety factor of $\eta = 1.26$ and a maximum von Mises stress of $214 MPa$, which is considered acceptable for the considered extreme loading scenario. The results validated the chosen material and design approach, proving that the LCA can withstand the applied forces while maintaining a lightweight structure.

However, further improvements could focus on refining the manufacturing process to enhance cost-efficiency and durability. Additionally, a fatigue analysis could be conducted to ensure long-term reliability under cyclic loading conditions. Exploring alternative materials with higher strength-to-weight ratios could also provide further weight reductions without compromising performance. Finally, real-world testing and validation would be beneficial to confirm the simulation results and to fine-tune the design based on empirical data.

Overall, this study demonstrates the effectiveness of an integrated design approach that combines kinematic optimization, structural analysis, and manufacturing feasibility, resulting in a well-balanced suspension component suitable for real-world applications.

4.8 Manufacturing techniques

In (Tab. 19) are reported the materials and production techniques chosen for the main components of the suspension. In particular, machining comprehends both turning and milling and is used to create the parts that must be in contact with some other components.

Part	Material	Process
Hub	Steel	Forging + Machining
Hub Carrier	Aluminium	Casting + Machining
Rim	Aluminium	Casting + Flow forming
Lower Arm	Aluminium	Casting + Machining

Table 19: Production processes and materials

5 Assignment 5: Wheel testing

5.1 Introduction

The aim of this assignment was to compute the fatigue life of an aluminium alloy wheel when subjected to the dynamic cornering fatigue test. During the test the wheel is clamped in the horizontal plane, while a rotating bending moment is applied on the wheel through an eccentric rotating mass.

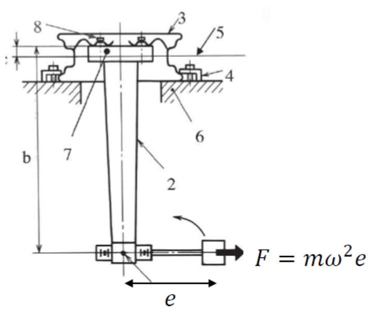


Figure 48: Fatigue test bench

During this test two orthogonal unitary bending moments are applied to the wheel, then the two resulting stresses are combined together and scaled up with a test bending moment M, in this way is possible to simulate the rotating bending moment effects.

The test bending moment is computed as in (Eq. 5.1), where R is the total tyre radius, d is the wheel inset, S is the accelerated test factor, F_v is the maximum vertical static load of the vehicle.

$$M = (\mu R + d) \cdot F_v S \tag{5.1}$$

The data used in the calculations are taken from the wheel used in the car presented in (Sec. 1.3.1), which is a 205/50 R17.

The stresses coming from the two different moments are combined depending on the angle α, which is varied from 0° to 360°, to simulate the rotation of the bending moment.

$$\sigma = M \cdot (\sigma_{dM_z} \cdot cos(\alpha) + \sigma_{dM_x} \cdot sin(\alpha)) \tag{5.2}$$

The sines fatigue criterion can be used to estimate fatigue life, where σ and τ are the alternating components of the stresses.

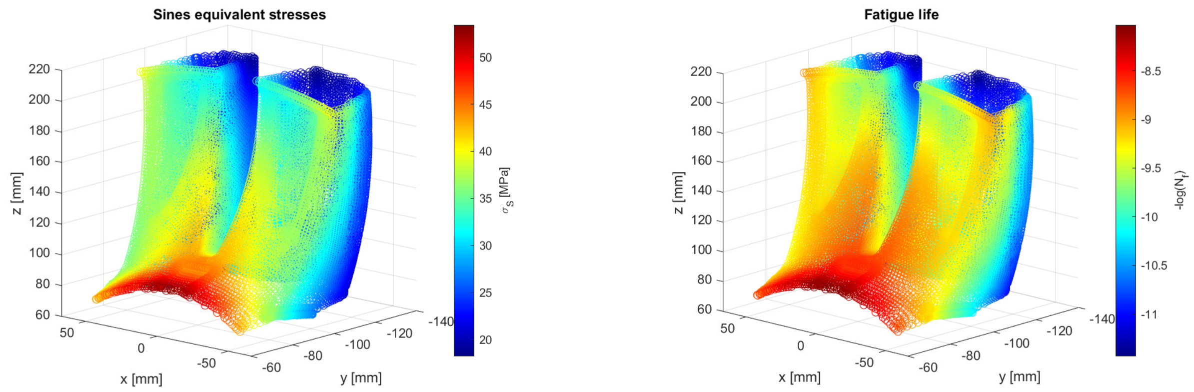
$$\sigma_S = \sqrt{\sigma_{x,a}^2 + \sigma_{y,a}^2 + \sigma_{z,a}^2 - (\sigma_{x,a}\sigma_{y,a} + \sigma_{x,a}\sigma_{z,a} + \sigma_{y,a}\sigma_{z,a}) + 3(\tau_{xy,a}^2 + \tau_{xy,a}^2 + \tau_{xy,a}^2)} \tag{5.3}$$

Then the Wohler's curve is used to compute then fatigue life of the wheel, where $\sigma_f' = 741.7584\,MPa$ and $b = -0.1371$.

$$N_f = \frac{1}{2}\left(\frac{\sigma_S}{\sigma_f'}\right)^{\frac{1}{b}} \tag{5.4}$$

5.2 Results

Below are shown the results of the analysis in terms of equivalent stress and fatigue life on two spokes of the wheel (the other spokes are equal for symmetry) (Fig. 49).
The outer side of the wheel appears to be relatively lightly loaded, at around 20 MPa, with a fatigue life of around 10^{11} cycles.



(a) Sines equivalent stresses

(b) Fatigue life

Figure 49: Simulation results

On the inner side instead there is a high concentration of loads, with equivalent stresses up to around 54 MPa, with a fatigue life of around 10^8 cycles, which, if considering a normal passenger vehicle, is around the actual expectancy life of a car. A wheel approximately makes 25 millions revolutions each 50000 km, meaning that with a usual life expectancy of 200000 km the wheel will make 100000000 revolutions, which is exactly 10^8 so the fatigue life of the wheel itself. This means that the wheel should be redesigned to have a longer fatigue life to avoid unwanted ruptures in the case the car travels more than 200000 km.

This stress concentration region could be caused by several reasons, such as the geometry of the rim and the distribution of the loads which come from the bolted connection. Moreover, a possible notch effect could be generated by the presence of the two-spokes connection: the combination of these and other possible effects could be the reason for the higher stresses experienced in that rim portion.

6 Drismi Experience

6.1 Introduction

On the 25 October 2024, Politecnico Di Milano gave us the opportunity to visit and test its driving simulator with the aim of better understanding the relations between road excitations, vehicle response and human perception in terms of comfort.
The DRISMI is capable of reproducing a total of six degree of freedom (three translations, pitch, roll and yaw motions). The focus of the experience is centred in noticing the differences between a 1 Hz excitation and a 15 Hz one.

6.2 Theoretical background

A theoretical explanation regarding the quarter vehicle model gave us the knowledge to better judge what we've felt during the experience. A car suspension behavior can be modeled as shown in (Fig. 50).
The system is a two degrees of freedom, which are the vertical displacements of the sprung and unsprung masses, which results in two resonances associated to the two bodies. A typical FRF of the vertical acceleration of the sprung mass, related to the car discomfort, was plotted in (Fig. 51), where the resonances are clearly visible: at $1 Hz$ the car body one, while at about $14 Hz$ the unsprung mass one.

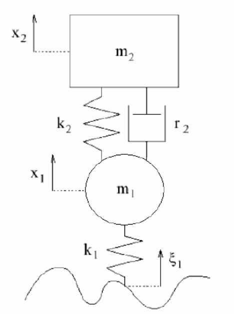


Figure 50: Quarter vehicle model

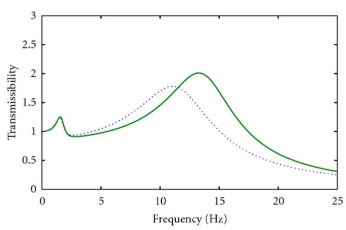


Figure 51: Discomfort frequency response

In (Fig. 52) the main human body organs resonances are shown. It's important to observe that many resonance frequencies are about 10 - 20 Hz while no resonances are positioned at 1 Hz (the minimum is about 5 Hz). We expect to perceive way higher excitations while being subjected to a 15 Hz frequency with respect to the 1 Hz case, which is instead approximately equal to the frequency related to human walking steps and heart beating, both situations at which the body is naturally used to sustain.

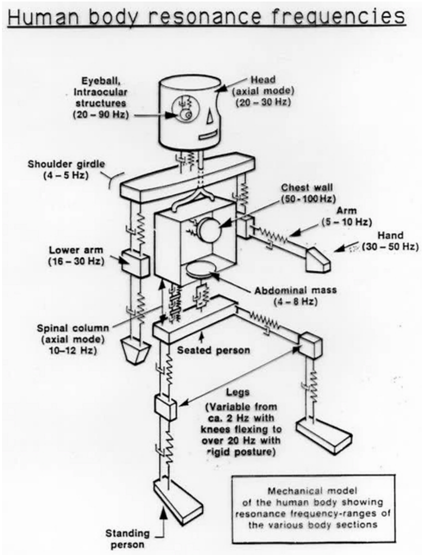


Figure 52: Human body resonances

6.3 Simulator experience

The experience consisted in sitting in the cockpit while it was moving.
At first two single frequency excitations where activated separately, one at 1 Hz, one at 15 Hz. Then the vehicle was made run over 5 speed bumps, simulating a series of Dirac-delta excitations, which resulted in more frequencies being excited simultaneously.
When running at 1 Hz, the car body mode motion was activated, but it did not feel much uncomfortable due to the fact that 1 Hz is more or less the frequency related to walking, and the human body is well capable of sustain it. Instead, the 15 Hz motion resulted in a very uncomfortable ride, because of some body organs resonances, like the jaw, resulting in the teeth clashing, which is not sustainable for a long period of time. When hitting the bumps, the main contribution that could be felt right away, was the one of 15 Hz. Instead, the contribution coming from the 1 Hz component was present but very lightly felt, only when the car was settling after having hit the bump. Reducing the peak of the second resonance of the system is fundamental for a car which aspires to be comfortable.

7 Capgemini seminar

7.1 Introduction

Capgemini is the largest management consulting company in Europe, working in the automotive, aerospace, railway, and defence field, but also in the growing industries such as the energy transition sector.
On the 13 December 2024, Capgemini gave us the opportunity to learn more about Functional safety, in particular with a focus on the ISO 26262 which is the standard for Functional safety related to the electrical and electronic systems sdopted in the automotive field.
The basic concept related to Safety are the definition of Hazard, Risk and Harm. Hazard is the potential source of harm caused by malfunctioning of an item. Risk is defined as the combination of the probability of occurrence of the harm and the severity of the harm. Harm is the physical injury or damage to the health of people or the environment.
Safety is defined as the freedom from unacceptable risk of physical injury. This definition implies that the risk of injury is not zero and is defined as a level of risk that is unacceptable.
We should distinguish between fault and failure. A fault is an abnormal condition that can cause an item to fail, while a failure is the termination of the ability to perform a task from an item. Failure can be divided in two main categories: Random hardware failure and Systematic failure. Random failure can occur unpredictably during the lifetime of an item and follow a probability distribution. Systematic failure occur in a deterministic way and can be eliminated only through a change in the design or in the processes.
An hardware fault tolerance can be defined as the hardware capability to fault resistance.
There are two main ways to reduce this risk, which are the following:

- Passive Safety
- Active Safety

A passive safety feature is a system that helps to reduce the effect of an accident and it does not work until it is called to action. An active safety feature works to prevent the accident. An example of these systems in the automotive field are the antilock-braking system and the adaptive cruise control.
A safety function is a function implemented by an electrical system that is intended to mantain the safe state of the equipment under control, with respect to a specific hazard. Each safety function has associated a SIL (Safety integrity level), which is a discrete level in the range between 1 and 4.
Based on the required SIL, you should reach proper values of failure rate, safe failure fraction and hardware fault tolerance on an hardware point of view and apply a set of rules to reduce the probability to introduce systematic failures from a software point of view.

The general safety development approach consists in 4 main step:

- Definition of the subject of the analysis
- The hazard analysis and risk assessment
- The safety function definition and the allocation of SIL
- Definition of the measure and method to follow according to the allocated SIL

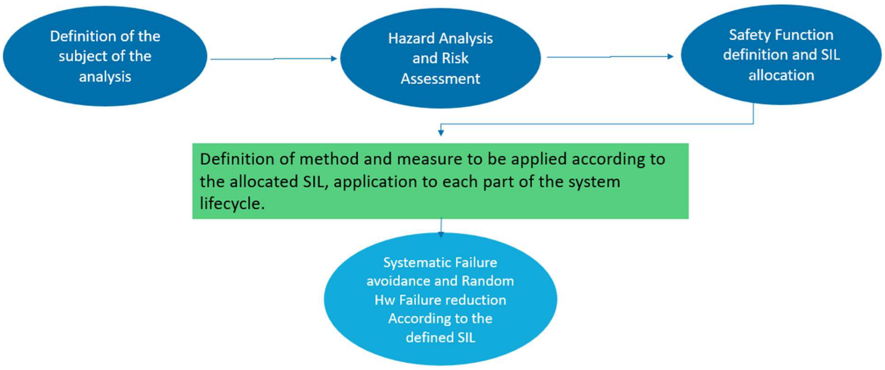


Figure 53: Safety development approach

7.2 ISO 26262

ISO 26262 is a provisional standard that provides methods to ensure safety compliance of an item, which leads to a great freedom in technological development.
The aim of the standard is to translate the results of the Activities into hierarchical requirements, providing traceability between them.
In this way both design and realization will fulfil the ISO 26262 compliance and the final product will have the characteristics enlisted in the design documents.
The standard always allow you to perform a tailoring on the activities: tailoring consists into deciding not to perform an activity, or not to use a specific method/measure. The only constraint consists into providing a robust rationale on why one decided to act so.

Figure 54: ASILs

ASIL refers to Automotive Safety Integrity Level and it is a risk classification system defined by the ISO 26262 standard for the functional safety of road vehicles. There are four ASILs identified by ISO 26262 : A, B, C, and D. ASIL A represents the lowest degree and ASIL D represents the highest degree of automotive hazard.
Factors that judge whether a risk is acceptable or not :

- Probability of exposure to hazardous events
- Severity of potential injuries
- Controllability of the situation

ASIL from A to D means that there is some level of non-acceptable risk in the system and particular Functional safety efforts are needed to raise the controllability of unwanted situations.

7.3 Failure mode and effect analysis

Failure mode and effects analysis (FMEA) is the process of reviewing as many components, assemblies, and subsystems as possible to identify potential failure modes in a system and their causes and effects. Generally, this kind of analysis can be done preventively, as part of the Preventive Reliability approach, i.e. the set of activities that allow to obtain the quality expected by the Customer and keep it in its lifecycle. This method is applied during the design phase development and its output is the prevention/resolution of issues as well as the creation of a database with the Lessons Learned.

Figure 55: Steps in FMEA approach

FMEA is a preventive analysis tool used to:

- identify potential failure modes of a new design item and their effects to the user
- identify the related potential root causes
- identify the critical design characteristics
- analyse the validation plan to detect the failures prior to the Start of Production
- define an effective corrective actions plan able to reduce the risk of each issue

In order to asses the design of an item, a block diagram of it should be created according to the following set of rules:

- define all elements of the component analysed
- highlight the interactions between the elements with internal/external physical/not physical interfaces
- define the perimeter of analysis
- highlight the responsibility of the analysis of the elements/interfaces identified

A risk matrix should be created, and on its basis the identification of critical parameters can be done, giving precise indications to the process industrialization and to invest the budget where it is really needed.

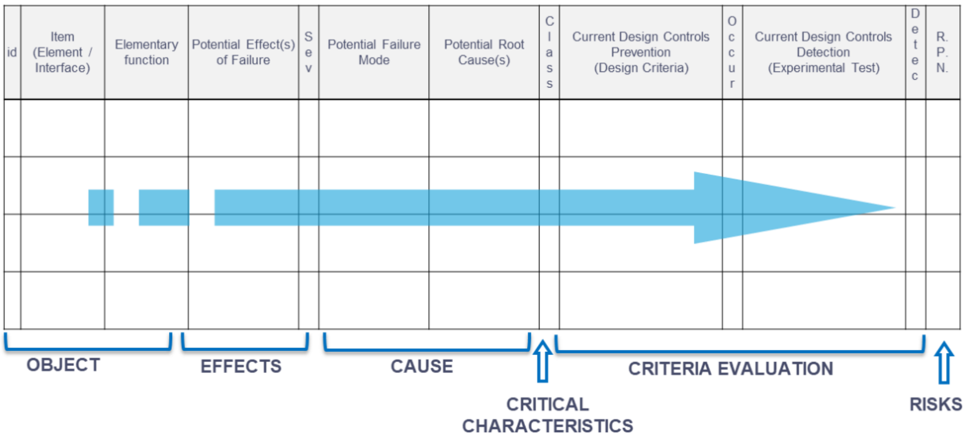


Figure 56: Risk Matrix

In the risk matrix the main columns are:

- Potential Effect Failure is a description of what it is perceived by the customer if the function is partially or completely not guaranteed.
- Severity (SEV) is a rating assigned to each effect of failure and indicates how serious the consequences of a failure mode are.
- The Detection is an index that indicates the probability of a failure to be detected and isolated by an operator.
- Potential Failure Mode are all the failure modes that can cause the effects identified.
- Potential Root Causes are the mechanisms that can lead to the failure analysed.
- The Risk priority Number (RPN) is calculated by multiplying the Severity, the probability of occurrence and the detection of a Failure.

The RPN indicates the priority in which action must be taken and can lead to the edit of an Action Plan.
Key Performance Indicator (KPIs) are indicators used for the tracking of the meetings scheduling, of the analysis progress status and of the corrective actions plan status.
Design-FMEA in all its forms is mostly required by automotive OEMs and Tiers with the aim of analysing new components or systems in terms of hardware robustness.

8 Cromodora seminar

8.1 Introduction

Cromodora is an Italian company, born in 1962, specialized in the production of magnesium alloy wheels for standard production and competition use. Cromodora is producing wheels through the following processes:

- Low-pressure die casting
- Flow Forming

8.2 Low-pressure die casting

Low-pressure die casting is a method of production that uses pressure to fill dies with molten aluminium. The holding furnace is located below the cast and the liquid metal is forced upwards into the cavity. The pressure is applied constantly to hold the metal in place within the die until it solidifies. Once the cast has solidified and cooled, the die is opened and the cast is removed.

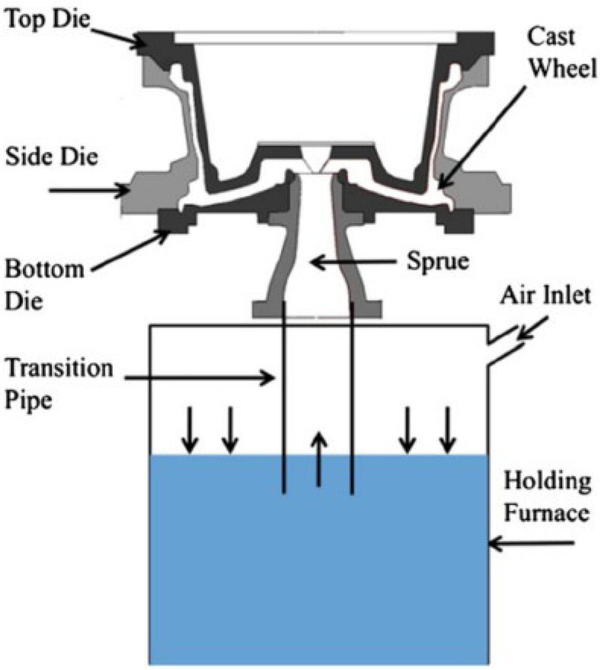


Figure 57: Low-pressure die casting

This is the most common manufacturing technology in the market regarding aluminium wheels, as they can produce good material characteristics at a reasonable cost.
After being casted, the wheels are x-rayed to asses the presence of imperfections and heat treated to improve the material behaviour.

8.3 Flow forming

Flow forming is one of the most advanced manufacturing technologies used to produce wheels, it involves the application of pressure to the inner barrel of the wheel while spinning and it is applied after casting. In this way the wheel is stretched up to the desired length and shape. This process improves the tensile strength of the aluminium, because it orients all the grains in the structure of the material and at the same time it reduces its porosity. In this perspective, the process shares similar properties to those found in the forging process. In this way, a much thinner and lighter wheels can be produced compared to casting only.

Figure 58: Flow Forming

8.4 Forging

Another process widely used to produce wheel is forging, which can be of two kind:

- Net forging
- Blank forging and machining

Net forging is for high volumes and low-cost forged wheels, in the range of cast wheels, but the variety of styles is very limited due to manufacturing limitations.
Blank forging and machining from solid allows to manufacture very elaborate styles, but the cost is an order of magnitude higher than all other processes. The advantage comes from overcoming the problem of draft angles and model extraction, in order to obtain a wide range of styles. These wheels are typically fit on super or hyper cars.

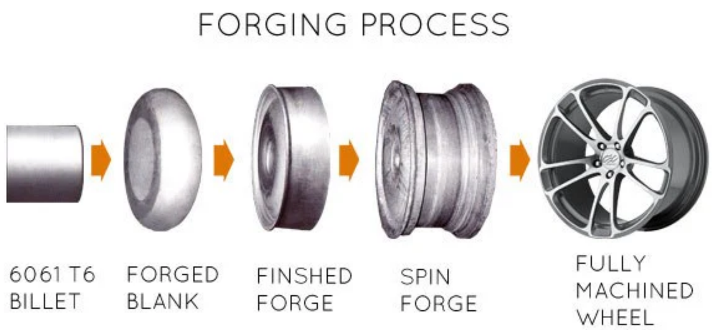


Figure 59: Forging

9 Pirelli testing track experience

9.1 Introduction

On the 29/11/2024, Pirelli gave us the oppurtunity to visit their testing track in Vizzola Ticino. This facility is mainly used for testing tires in wet condition and to perform test on noise characteristic, in fact the track is equipped with an irrigation system that can constantly supply a layer of water on the proving ground.
During our visit three type of tests were performed:

- Steering pad
- Aquaplaning
- Handling

All these tests have been performed under wet road conditions with a rear wheel drive sedan.

9.2 Steering pad

The steering pad test consists in driving the car keeping a circular trajectory with fixed radius, while constantly increasing the speed. This is done in order to characterize the handling of the car.

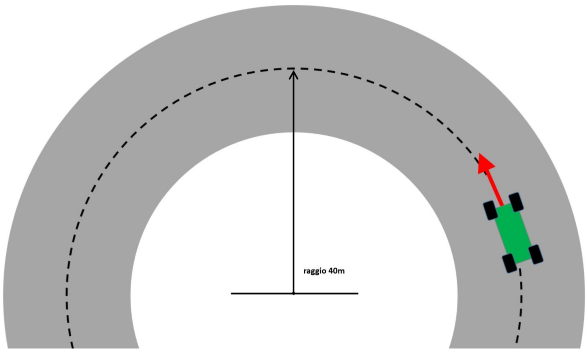


Figure 60: Steering pad manoeuvre

At low velocity the vehicle exhibit an understeering behaviour, which is what is expected from a road car due to safety reasons. In fact an understeering car can be considered safer, easier to drive and more stable and controllable for a non-skilled driver.
At the very end of the test, the driver has been capable of induce an oversteering behaviour through the use of the accelerator pedal by exceeding the friction limit on the rear tires, which is lowered by the wet condition of the track.
In order to maintain the control of the vehicle, the driver had to apply a counter-steering action, while, at the same time, he had to continuously press and release the throttle to maintain and control the oversteering behaviour.

9.3 Aquaplaning

The Aquaplaning test were performed in order to asses the behaviour of the tires running over an 8 mm layer of water.
The test has been repeated at two different speeds, and the behaviour of the vehicle changed completely between the two tests.
In fact, while running over the layer of water at a relatively low speed, the vehicle would react in a predictable way, because the tires have the capacity of evacuate enough water to remain in contact with the ground underneath. With an increased speed, the tires do not have the capacity to evacuate enough water, and a layer of water keeps the wheel separated from the ground, making the wheels completely loose their steering capability: while maintaining the wheel spinning through the accelerator pedal, no lateral forces could be transmitted from the tyres, making the car run straight ahead even if the steering wheel is completely turned.

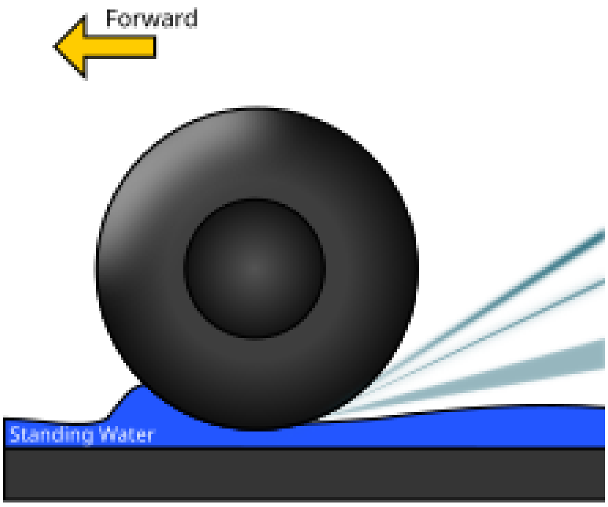


Figure 61: Aquaplaning

9.4 Handling

The handling test consisted in driving around the testing track in order to verify and asses its behaviour, while keeping an high level of lateral acceleration, and as a consequence by making it work outside the linear range, to test the tyre behaviour in many different highly demanding situations, such as high speed and low radius cornering, low grip situation, slalom and so on.

10 Brembo exercise

10.1 Introduction

The braking system of a motorbike was analysed, computing forces acting both on the vehicle level and on the system level, analysing also the braking force to be applied by the driver on the lever.

The model describing the motorbike (Fig. 62) is based on simple equilibrium equations (Eq. 10.1):

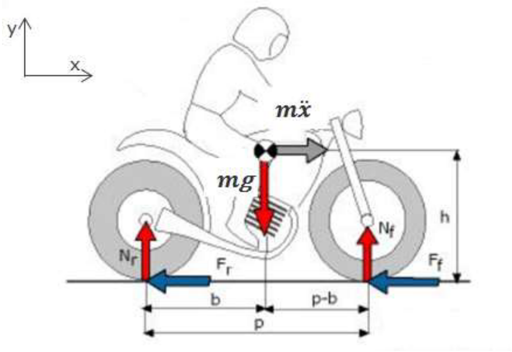


Figure 62: Motorbike model

$$\begin{cases} m\ddot{x} - F_r - F_f - F_{aero} = 0 \\ -mg + N_r + N_f = 0 \\ -(F_f + F_r)h - N_r b + N_f(p-b) = 0 \end{cases} \tag{10.1}$$

The data used in the calculations are reported in (Tab. 20).

m	217 kg
h	677 mm
p	1382 mm
b	655 mm
μ_{tyre}	0.8

Table 20: Vehicle data

Meanwhile the data used to compute the aerodynamic forces when considered is reported in (Tab. 21).

v	200 km/h
C_d	0.6
A	0.4 m^2
ρ	1.2 kg/m^3

Table 21: Aerodynamic forces data

10.2 Requests

The requests are split in two parts: one focusing on the overall vehicle behaviour under different braking conditions and one where the actual braking system is analysed.

10.2.1 Motorbike behaviour

1. Static load distribution: In static conditions the horizontal forces of (Eq. 10.1) are not present, therefore solving for N_f and N_r the equations read:

$$\begin{cases} N_f^{st} = m_{tot} g \frac{b}{p} = 1334.36\,N \\ N_r^{st} = m_{tot} g \frac{p-b}{p} = 1481.11\,N \end{cases}$$

2. Deceleration at flipping limit without aerodynamic effects: When considering the bike at its flipping limit the rear vertical force N_r is null, therefore without considering aerodynamic forces (Eq. 10.1) is solved for the deceleration $\ddot{x}$, resulting in:

$$\ddot{x}_{max} = g\frac{p-b}{h} = 10.53\,m/s^2$$

3. Maximum deceleration achievable without aerodynamic effects: By considering the same friction coefficient μ_{tyre} between front and rear wheel, the system of equations (Eq. 10.1) is simplified down to find that the maximum achievable deceleration is described by the Coulomb's friction law:

$$\ddot{x} = \mu_{tyre}\, g = 7.85\,m/s^2$$

4. Brake repartition: Analysing the previous case and solving for F_f and F_r is possible to compute the brake repartition as

$$\begin{cases} \frac{F_f}{F_{tot}} = \frac{b+\mu h}{p} = 86.58\% \\ \frac{F_r}{F_{tot}} = \frac{p-b-\mu h}{p} = 13.42\% \end{cases}$$

5. Maximum deceleration using either only front brake or only rear brake: Considering the same model described before it is possible to derive the maximum achievable deceleration utilizing only the front or the rear brakes, still without considering aerodynamics effects:

$$\begin{cases} \ddot{x}_f = g\frac{p\cancel{\mu_r}+b(\mu_f-\cancel{\mu_r})}{p+h(\cancel{\mu_r}-\mu_f)} = 6.12\,m/s^2 \\ \ddot{x}_r = g\frac{p\mu_r+b(\cancel{\mu_f}-\mu_r)}{p+h(\mu_r-\cancel{\mu_f})} = 2.97\,m/s^2 \end{cases}$$

6. Maximum deceleration achievable considering aerodynamic effects: The equation utilised in the point 3 can be reformulated to account for the aerodynamic effects, considering the data provided in (Tab. 21):

$$\ddot{x}_{max} = g\frac{p-b}{h} + \frac{1}{2}\frac{\rho C_d A v^2}{m_{tot}} = 12.09\,m/s^2$$

The aerodynamic forced will help the bike decelerate more as they are directed opposite to the bike direction effectively slowing it down.

10.2.2 Braking system design

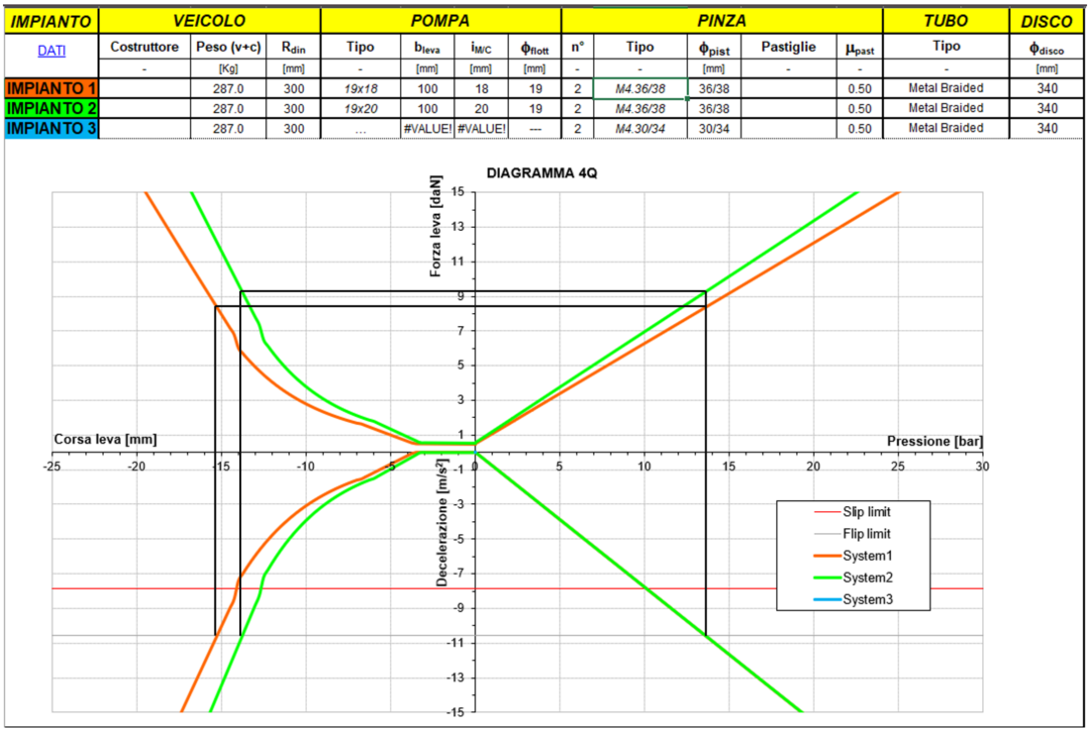
The same vehicle described in (Sec. 10.1) was utilised for this analysis, with the addition of more specific data regarding the braking system.

Calipers	36/38
D_{disk}	340 mm
μ_{pad}	0.5

Table 22: Braking system data

All the requests were carried out by analysing the so called 4Q diagram showed in (Fig. 63), which was provided by Brembo.

1. System pressure at flipping limit: By drawing a vertical line where the deceleration hits the maximum flipping limit is possible to find the system pressure: $p_{system} \approx 13.6\,bar$.
2. Rider force with 19X18 master cylinder: By continuing the previous vertical line up to when it intersects the line of the braking force at the lever with this cylinder (red line) is possible to find the force that the driver's hand must apply to brake at flipping limit: $F_{lever} \approx 84N$.
3. 19X20 master cylinders By changing the master cylinders to the new specification (green curve) means that the geometry of the actual lever has changed, since the positions of the joints moved. In this particular case, respect to the previous one, it resulted in a higher force required to the driver, $F_{lever} \approx 93,N$ but at the same time less movement of the lever is required to reach the same flipping deceleration limit.



IMPIANTO	VEICOLO			POMPA				PINZA					TUBO	DISCO
DATI	Costruttore	Peso (v+c)	R_{din}	Tipo	b_{leva}	$i_{M/C}$	ϕ_{flott}	n°	Tipo	ϕ_{pist}	Pastiglie	μ_{past}	Tipo	ϕ_{disco}
	-	[Kg]	[mm]	-	[mm]	[mm]	[mm]	-	-	[mm]	-	-	-	[mm]
IMPIANTO 1		287.0	300	19x18	100	18	19	2	M4.36/38	36/38		0.50	Metal Braided	340
IMPIANTO 2		287.0	300	19x20	100	20	19	2	M4.36/38	36/38		0.50	Metal Braided	340
IMPIANTO 3		287.0	300	...	#VALUE!	#VALUE!	---	2	M4.30/34	30/34		0.50	Metal Braided	340

Figure 63: 4Q Diagram